MSc Physics & Astronomy

General Physics

Master Thesis

An analytic application of the Fluctuation-Dissipation Theorem in Alzheimer's Disease using the Linear Hopf Model

by

Melle Bolding

Supervisors:	Dr. Gustavo Patow (UDG, UPF)
	Dr. Gustavo Deco (UPF)
	Dr. Greg Stephens (VU)
Second Examiner:	Dr. Clélia de Mulatier (UvA)
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Abstract

Cognitive function in healthy brains depends on sustained non-equilibrium dynamics that indicate hierarchical information processing across various brain networks. In Alzheimer's disease (AD), this organization is disrupted, leading to more equilibrium-like dynamics and a flatter hierarchy of information integration. In this study, we have applied both a model-free and analytical model-based implementation of the fluctuation–dissipation theorem to resting-state fMRI data to quantify deviations from equilibrium in brain dynamics. We have observed a significant reduction in hierarchical organization in AD patients at the whole-brain level, as well as within the salience, limbic, and default mode networks, which are associated with AD pathology and cognitive symptoms. Notably, the spatial distribution of non-equilibrium deviations showed only minimal correlation to amyloid-beta and tau protein depositions, suggesting that dynamical alterations are not a direct reflection of local AD pathology. Finally, using a machine learning approach, we demonstrate that non-equilibrium metrics provide complementary information to neuropathological measures and improve classification performance in early-stage AD. Together, these results provide insight into how functional alternations relate to pathology and cognitive decline in AD.

Contents

List of Figures

Nomenclature

AD	Alzheimer's Disease
ASL	Arterial Spin Labeling
BOLD	Blood Oxygenation Level-Dependent
CBF	Cerebral Blood Flow
CMRO_2	Cerebral Metabolic Rate of Oxygen consumption
CSF	Cerebrospinal Fluid
COV(τ**)**	(τ-delayed) covariance matrix
DTI	Diffusion Tensor Imaging
DW-MRI	Diffusion-Weighted Magnetic Resonance Imaging
EC	Effective Connectivity
EEG	Electroencephalography
FC	Functional Connectivity
FDT	Fluctuation–Dissipation Theorem
FWHM	Full Width at Half Maximum
GM	Grey Matter
HC	Healthy Control
HRF	Hemodynamic Response Function
IV	Integral Violation
MCI	Mild Cognitive Impairment
MEG	Magneto-Encephalography
MRI	Magnetic Resonance Imaging
PET	Positron Emission Tomography
rs-fMRI	resting-state functional Magnetic Resonance Imaging
RSN	Resting-State Network
SC	Structural Connectivity
fMRI	functional Magnetic Resonance Imaging
TTI	Time-Translation Invariant
TRS	Time-Reversal Symmetry
WM	White Matter
WBM	Whole-Brain Modeling
SUVR	Standardized Uptake Value Ratio
CL	Centiloids
RF	Random Forest
SVM	Support Vector Machine

1 Introduction

One of the most elusive phenomena in nature is the emergence of simple, coherent behavior from extremely high-dimensional dynamics. The human brain is a remarkable example of such a system. Comprising over 86 billion neurons with intricate non-linear interactions, the brain expresses highly complex activity across temporal and spatial scales. Yet, human behavior is strikingly structured and, for the most part, predictable. Explaining how such ordered macroscopic behavior arises from vast microscopic complexity remains one of the central endeavors in neuroscience, with at its heart the question of how neural structure gives rise to neural function.

Alzheimer's disease (AD) presents a particularly interesting case for investigating this structure–function relationship. AD is marked by the progressive accumulation of amyloid-beta plaques and tau neurofibrillary tangles, which disrupts synaptic signaling and leads to network-scale dysfunction [1]. Although the spatial progression of pathology, from early protein aggregations in the medial temporal structures to later neocortical spread, is well characterized, the field still lacks a mechanistic explanation of how these structural changes translate into the functional impairments observed in clinics, such as episodic memory loss and global cognitive decline [2]. Furthermore, these functional impairments do not emerge uniformly; patients often preserve substantial cognitive capacities for years. This suggests the presence of compensatory mechanisms that, at least temporarily, act as buffer against structural alternations [3].

In an attempt to gain insight into the underlying dynamics, we turn to a concept from statistical physics. The fluctuation-dissipation theorem (FDT) links the spontaneous fluctuations of a system to its linear response to external perturbations; dissipation. As the relationship only holds exactly at detailed balance equilibrium, it allows for an estimation of the violation of this equilibrium [4], [5]. In biological systems, deviations from detailed balance provide signatures of non-symmetrical processes, including directional information flow, hierarchical organization, and sustained entropy-producing activity [6]. Applying this framework to the brain thus provides a way to quantify its distance from detailed balance equilibrium and enables regional insights into how AD pathology affects the system's dynamical characteristics.

As the FDT framework is designed to act upon steady-state dynamics, it is essential to robustly characterize the system dynamics. Whole-brain models provide just that by embedding empirically measured data from structural and functional neuroimaging modalities, such as MRI and PET, into a system of coupled, stochastic oscillators [7]. In this thesis, we apply the linear Hopf model, a linearized version of the Hopf bifurcation normal form that remains analytically tractable while capturing the transition between noise-driven and oscillatory regimes [8]. Crucially, this model enables an accurate estimation of the stationary statistics, which allows for an analytical implementation of the FDT. Next to this model-based approach, we use a model-free application that involves a Wiener filter to split deterministic and stochastic parts of the empirical time series data [9].

Applying this framework to a clinical resting-state fMRI dataset, ADNI [10], we find that AD progression is associated with a reduction in distance from detailed balance equilibrium: AD brain dynamics shift toward more equilibrium-like behavior. A possible interpretation is that the brain loses its ability to sustain the non-equilibrium dynamics that support healthy cognitive function as AD progression affects its structural integrity. Moreover, we observe that changes in many non-equilibrium regions do not correlate with amyloid-beta and tau protein deposition patterns. Nonetheless, most regions with significant change in equilibrium violation are located in sections associated with the manifestation of AD symptoms, such as the posterior cingulate and precuneus cortex in the default mode network [11].

In general, these results align with a growing body of evidence portraying AD as a disease of information integration degradation and the distance from equilibrium as a general proxy of cognitive hierarchy. Previous work in AD has shown reductions in irreversibility [12] and metastability [13], and disruptions in the excitation-inhibition balance [14]. Additionally, a previous FDT study in AD has presented a similar decline in distance from equilibrium [9]. Other FDT papers have shown that the distance from equilibrium increases in both awake and task states as compared to sleep and rest states, respectively [15], [16]. Overall, the main results of this study are in support of the notions that (1) AD is characterized by a progressive loss of non-equilibrium dynamics, and (2) the distance from equilibrium metric is positively related to cognitive complexity.

Alzheimer's disease

Alzheimer's disease is a progressive neurodegenerative disorder and the most common cause of dementia. The disorder is characterized by a decline in episodic memory function followed by impairments in executive function, language, visuospatial abilities, and eventually global cognition [2]. Disease progression is slow and heterogeneous, with substantial variability between patients in symptom onset and rate of decline. Notably, patients often retain significant cognitive function for extended periods despite ongoing neuropathological changes, indicating that functional impairment does not arise as a direct consequence of structural alternations [3].

Pathologically, AD is defined by the accumulation of extracellular amyloid-beta plaques and intracellular neurofibrillary tangles composed of hyperphosphorylated tau proteins (Fig. 1.0.1) [17]. These aggregations of misfolded proteins follow a characteristic spatial progression pattern, with early involvement of the medial temporal lobe structures and subsequent neocortical spread [18]. This spread is accompanied by loss of synapses, neuroinflammatory responses, and eventual neurodegeneration [1]. While the spatial deposition pattern is well documented, its relationship to cognitive decline is not straightforward, as protein burdens correlate only moderately with clinical symptoms, particularly in early disease stages [2].

This mismatch complicates both diagnosis and clinical course predictability. Concretely, neuropathological markers of AD can be present for years before the emergence of clinical symptoms, and in some patients substantial pathology does not translate into noticeable functional impairments [3], [19]. As a result, structural measures alone are insufficient to explain AD, motivating a turn towards functional and dynamical descriptions of AD progression.

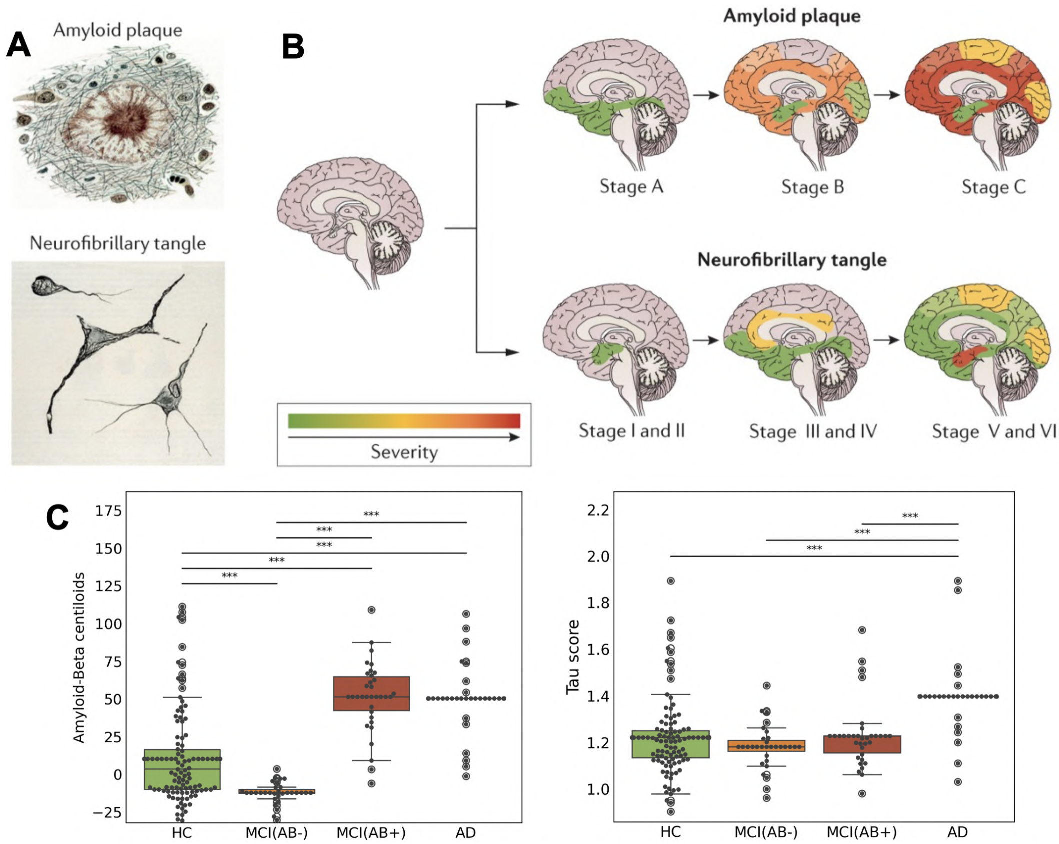

Figure 1.0.1: Alzheimer's disease pathology. Accumulation of extracellular amyloid-beta plaques and intracellular neurofibrillary tangles composed of hyperphosphorylated tau proteins are hallmarks of AD pathology (**A**). Spatial depositions of these proteins follow an established pattern as the disease progresses (**B**). Subject averages for amyloid-beta and tau values of the PET dataset used in this study are shown for the groups HC, MCI(AB-), MCI(AB+), and AD (**C**). Statistical significance between grouped subject averages is determined by the Wilcoxon rank-sum test; ***: p-value < 0.001. Figures **A** and **B** were adapted from [18].

Alterations in network-scale brain dynamics are a prominent feature of AD. Resting-state fMRI studies show disruptions in several functional networks, most prominently in the default mode and salience networks, which play a important roles in memory and internally directed cognition [20]. However, these disruptions are stage-dependent. Early AD is often associated with elevated neuronal activity in memory-related regions [21], whereas later stages show widespread network breakdown and hypoconnectivity [11]. The interpretation of early hyperactivity remains disputed. Some view hyperexcitability as part of a compensatory mechanism that sustains cognitive function in the face of structural alternations. Others argue that it reflects network instability driven by an excitation–inhibition imbalance, contributing to synaptic stress and accelerating disease progression [22]. In either case, these observations indicate active reorganization of brain dynamics during AD progression, rather than uniform degradation of function.

In general, network disruptions indicate a disturbance in information integration. The human brain relies on a delicate balance of directed, non-reciprocal signaling to process information. However, recent studies show reduced irreversibility [12] and metastability [13], along with a disrupted excitation–inhibition balance [14] in AD. This suggests that brain dynamics in Alzheimer patients are moving toward a flatter hierarchy with more symmetric interactions across brain regions. By analyzing how far brain dynamics are removed from equilibrium, we can substantiate this hypothesized hierarchical disturbance in AD. Furthermore, the approach presented in this thesis might allow us to connect structural alternations with functional changes such as disturbed information integration in the Alzheimer brain.

Related work

Since the 2010s, thermodynamic principles have been introduced to computational neuroscience to describe whole-brain dynamics, linking structure with function and information processing [7]. However, the fluctuation-dissipation framework was only introduced to whole-brain neuroscience in 2023 [15]. In this study by Deco et al., the FDT framework was used to describe the differences in deviation from equilibrium between awake/sleep and rest/task brain states. They used the standard Hopf model to simulate the brain dynamics and then computed a spin-system version of the FDT. They found that the deviation from FDT equilibrium decreases significantly in deep sleep compared to awake and in rest compared to any of the seven tasks.

Around a year later, Monti et al. [16] presented their results on non-equilibrium deviations in awake/sleep. Although they used the same brain model, they computed a different version of the FDT. They used the integral violation, a variant from which the one applied in this thesis was derived. Using this FDT metric, they were able to show an even more significant decrease of distance from equilibrium in deep sleep compared to wakefulness.

Around the same time, a study by Patow et al. [9] applied the integral violation FDT to Alzheimer's disease. In this case, they used both a novel model-free approach and a dynamic mean field model-based implementation. Even though the paper is still in preprint, the results show that after an initial increase from HC to MCI, the distance from equilibrium decreases overall from HC to AD. Notably, the authors relate the initial increase in the MCI group to neuronal hyperexcitability, a phenomenon associated with the loss of inhibitory neuron function in AD [21].

More recently, Acero-Pousa et al. [23] published a study that used a similar approach to [15] in order to study schizophrenia. Compared to healthy controls, schizophrenic patients displayed an increase in distance from equilibrium. Furthermore, the limbic, subcortical, default mode, and control networks were identified as main drivers of FDT equilibrium deviations. Also, using a support vector machine classifier, the authors were able to show increased classification performance with FDT metrics as compared to functional or effective connectivity measures.

In conclusion, these four whole-brain FDT studies portray generally consistent results. They support the notion that the distance from equilibrium is positively related to cognitive complexity and increased organizational hierarchy. Namely, both [15] and [16] show that the distance from equilibrium is increased in wakefulness as compared to deep sleep. Furthermore, results in [15] also indicate that the distance from equilibrium is larger in tasks than in rest. Similarly, in [9], going from HC to AD decreases the distance from equilibrium while hyperexcitablility in MCI slightly increases it. Lastly, [23] show that in schizophrenia, a condition related to altered hierarchical organization [24], the deviation from FDT equilibrium significantly increases.

Outline of thesis

The structure of this thesis is as follows. In Chapter 2, we start by providing a select introduction to the field of Whole-Brain Modelling (WBM). This includes a section on neuroimaging modalities, such as magnetic resonance imaging (MRI) and positron emission tomography (PET) methods (Sec. 2.1), a section on analysis tools, such as connectivity measures and network metrics (Sec. 2.2), and lastly, a section on local brain dynamics, such as the Stuart-Landau model and its linearization (Sec. 2.3).

In the next chapter, Chapter 3, we introduce the fluctuation-dissipation theorem (FDT). We briefly explain the basic principles, the assumed Langevin dynamics and the integral violation metric (Sec. 3.1). Next, we describe the model-free implementation, which computes the violation of the FDT by approximating non-equilibrium steady-state dynamics and using the Wiener filter (Sec. 3.2). To close the chapter, we introduce a novel model-based analytical FDT framework (Sec. 3.3).

In Chapter 4, we turn to the main results of the thesis. First, we present the subject and parcel averages of the non-equilibrium metric for the healthy control (HC), mild cognitive impairment (MCI), and Alzheimer's disease (AD) groups (Sec. 4.1). Then, we show some results on the connection between the non-equilibrium metric and AD pathology (Sec. 4.2). In the next section, we implement a hybrid cross-validation classifier that enables the identification of important features (Sec. 4.3). Finally, we discuss the presented results, place them in the context of previous work, and mention some of the limitations (Sec. 4.4).

In the final chapter, Chapter 5, we summarize our findings and make recommendations for possible directions of future studies. In the appendix, we specify details of the dataset, further discuss the BOLD effect, expand on exact fitting procedure and model parameters, introduce the Wilcoxon rank-sum and Monte-Carlo permutation statistical tests, present the Duhamel derivation, and show additional figures.

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2

Modalities, Observables, and Coupled Oscillators

This thesis employs two distinct approaches to compute the same non-equilibrium metric. The main difference between these techniques is the use of a brain model. To provide the necessary foundation for this methodology, this chapter offers a pragmatic introduction to the field of Whole-Brain Modeling (WBM). In Sec. 2.1, we start by describing the data acquisition phase and briefly introduce two neuroimaging modalities: magnetic resonance imaging (MRI) and positron emission tomography (PET). Next, in Sec. 2.2, we explain some prevalent analysis tools, such as connectivity measures and network metrics. Finally, in Sec. 2.3, we turn our attention to a particular model of local brain dynamics. Specifically, we describe the Stuart-Landau model and its linear approximation. Readers already aware of these general modalities and analysis methods are encouraged to skip ahead to this final section.

2.1. Modalities

This section outlines the data acquisition phase. In this study, we use MRI and PET data collected by the Alzheimer's Disease Neuroimaging Initiative (ADNI). Further details on this dataset can be found in Appendix A.1. The acquisition stage is usually performed at hospitals or medical research centers and consists of the recording of the brain's internal anatomy or activity during a set period.

Often, the initial step includes acquiring the structural connectivity (SC) through diffusion neuroimaging techniques such as diffusion-weighted magnetic resonance imaging (DW-MRI) and in particular, diffusion tensor imaging (DTI). Additionally, PET scans are used to gather information on chemical depositions and molecular concentrations. In addition to the static structural and molecular data, we use data acquired through functional magnetic resonance imaging (fMRI) to capture active brain dynamics. After the data acquisition stage, the preprocessing steps are performed. These steps include denoising, artifact-removal, and normalization among others. Finally, the data is grouped into nodes according to criteria defined by a parcellation. In this study, we used the Schaefer-400 parcellation [25]. For further details, we again refer to Appendix A.1.

2.1.1. Magnetic Resonance Imaging

Magnetic resonance imaging is a non-invasive medical imaging technique that exploits the nuclear magnetic resonance properties of atomic nuclei, usually hydrogen protons in water and fat, to generate high-resolution images of biological tissue. When placed in a strong external magnetic field, the nuclear spins align with the field and can be excited by radiofrequency (RF) pulses. After excitation, the spins relax back to thermal equilibrium through two fundamental processes: longitudinal (T_1) and transverse (T_2) relaxation. The emitted RF signals, due to relaxation, are detected by receiver coils and spatially encoded using magnetic field gradients, allowing for the reconstruction of three-dimensional images [26].

The contrast in MRI images arises from tissue-dependent differences in proton density and relaxation times. This enables soft-tissue differentiation, in particular between gray matter, white matter, and cerebrospinal fluid. By choosing the pulse sequences and gradient encoding appropriately, MRI can be made sensitive to not only anatomical structure, but also to physiological function. This versatility forms the basis for modalities such as diffusion-weighted MRI (DW-MRI) for probing structural connectivity, and functional MRI (fMRI) for mapping brain activity via the hemodynamic response [27].

Diffusion-weighted MRI
Diffusion-weighted MRI is an MRI technique that probes tissue microstructure by measuring the diffusion of water molecules. Because water diffusion is constrained by cellular membranes, axonal fibers, and myelin sheaths, DW-MRI provides indirect information on the underlying tissue organization. Diffusion tensor imaging extends DW-MRI by acquiring diffusion measurements along multiple gradient directions, allowing the diffusion process within each voxel to be described by a tensor model. This enables the estimation of diffusion magnitude and directionality, providing measures of white matter fiber orientation, density, and integrity. For these reasons, DTI is widely used to study white matter microstructure and structural connectivity (Sec. 2.2.1) [27].

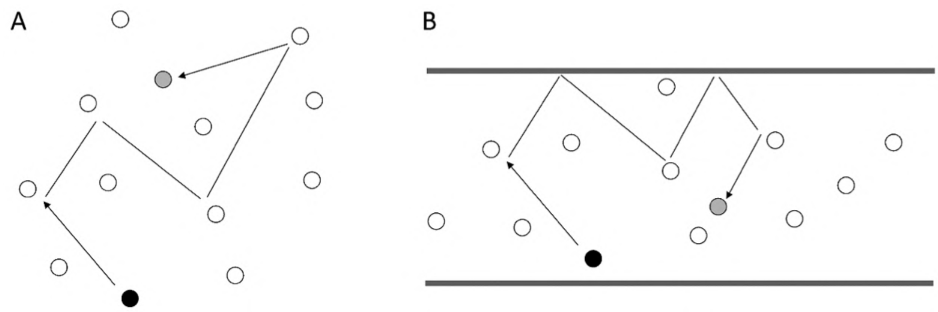

Figure 2.1.1: Sketch of the diffusion process, commonly described as random walk movement arising from thermal motion. We present (A) the unrestricted and (B) the boundary restricted case. In both cases, a molecule is initially represented as a black dot and through interactions ends up at the position of the grey dot. Adapted from [28].

Water diffusion arises from thermal motion and can be described as a random walk (Fig 2.1.1). The average net displacement $\langle d \rangle$ over a time interval Δt follows Einstein's diffusion relation,

$$\langle d \rangle = \sqrt{2D\Delta t}, \tag{2.1.1.1}$$

where D denotes the apparent diffusion coefficient. Tissue boundaries restrict molecular motion, leading to changes in D that can be measured using diffusion-sensitizing magnetic field gradients. A pair of equal and opposite gradients is applied; signal cancellation occurs only for stationary spins, while diffusion-induced motion causes phase dispersion and signal attenuation. The resulting MR signal intensity S is given by,

$$S = S_0 e^{-bD}, \tag{2.1.1.2}$$

where S_0 is the unweighted signal and b the diffusion-weighting factor. In white matter, diffusion is anisotropic, with higher diffusivity along fiber tracts than perpendicular to them. By applying gradients in multiple orientations, DTI exploits this anisotropy to infer tract orientation and organization [28].

Functional MRI and BOLD imaging

Next to mapping statical brain structure, MRI allows for the imaging of brain function using functional MRI methods. In particular, by using the blood oxygenation level-dependent (BOLD) effect, BOLD-fMRI maps brain activation patterns by detecting changes in blood flow and oxygenation levels, i.e., the hemodynamic response [27]. Although the term fMRI often coincides with BOLD imaging, the concept of functional imaging using MRI has wider applicability.

Zooming in on BOLD-fMRI, the signal arises due to two factors. First, through a biophysical mechanism: when hemoglobin, the blood molecule that carries oxygen, loses its oxygen, it becomes deoxyhemoglobin, a more paramagnetic molecule [29]. In contrast to diamagnets, paramagnets distort the local magnetic field and cause the induced MR signal to decay faster. This difference of the blood's magnetic susceptibility can be detected by MRI scanners [30]. However, only when this biophysical observation is combined with a second, physiological occurrence, does it become a relevant method for monitoring brain function.

All forms of tissue, including neuronal tissues, require oxygen to function. However, when local neuronal tissue activity increases, the blood flow to this area (CBF, cerebral blood flow) increases much more than the oxygen consumption (CMRO_2, cerebral metabolic rate of oxygen consumption); in other words, the vascular system overcompensates. Paradoxically, this causes more oxygenation in the local venous blood, despite the increase in oxygen uptake by the neuronal area. Together, these factors bring about a local increase in MR signal due to reduced oxygen depletion during increased neuronal activity; the BOLD effect [27].

Note that the BOLD effect is not a direct measure of neural activity. It arises from the metabolic activity that accompanies neural activity. To describe the relations between BOLD, metabolic activity, and neural activations, we need to tread into more detail, thus for the interested reader, we refer to Appendix A.2.

2.1.2. Positron Emission Tomography

Positron emission tomography is a neuroimaging technique that enables the in vivo quantification of biochemical processes through the use of radioactive tracers. After intravenous injection, tracers bind to specific molecules. Their radioactive decay leads to positron–electron annihilation events that produce pairs of gamma photons which are then detected and reconstructed into three-dimensional maps of tracer concentrations.

In Alzheimer's disease, PET is a used for visualizing pathological protein accumulation, in particular amyloid-beta plaques and tau neurofibrillary tangles. Amyloid-beta plaques are commonly imaged using tracers such as ^{11}C-PiB or ^{18}F-florbetapir, while tau tangles are identified using ^{18}F-flortaucipir [31]. Quantification is typically performed using the standardized uptake value ratio (SUVR),

$$\mathrm{SUVR} = \frac{C_{\mathrm{ROI}}}{C_{\mathrm{ref}}}, \tag{2.1.2.1}$$

where C_{ROI} and C_{ref} denote the tracer concentrations in the target and reference regions, respectively. Moreover, the reference region is assumed to be relatively free of the targeted molecules, usually the cerebellum is used for this.

Notably, PET scans capture slow pathological processes rather than fast neural dynamics [31]. In this thesis, PET data is therefore not used to study dynamic brain activity, but rather to provide static measures of the tau and amyloid-beta burden that contextualize the functional findings.

2.2. Observables

Where the previous section aimed to provide some insight into various neuroimaging modalities, this part describes several prevalent data structures derived from those techniques, often referred to as observables. In particular, we briefly touch upon connectivity measures such as the structural connectivity (SC), the functional connectivity (FC), and the delayed covariance matrix (COVτ). Next, we describe more advanced network-level observables such as the effective connectivity (EC) and the resting-state networks (RSNs).

2.2.1. Connectivity Measures

Connectivity plays a central role in contemporary neuroscience. From early on, the central nervous system has been conceptualized as a network of interconnected nodes, with considerable emphasis on the study of its wiring. Historically, this information was primarily derived from anatomical studies and tract-tracing experiments. However, with the emergence of modern neuroimaging, connectivity is no longer considered only in structural terms but also in dynamical patterns. In this part, we describe the structural connectivity, the functional connectivity and the delayed covariance matrix.

Structural Connectivity
Structural connectivity refers to the connectome of the brain, a map that describes the set of white matter tracts linking brain regions. It is often represented as an adjacency matrix where each element S_{ij} quantifies the strength of the connection between regions i and j.

Our understanding of SC originates from invasive tract-tracing studies in non-human primates [32], providing the first maps of between region connections in the brain. Later, the advent of diffusion-weighted magnetic resonance imaging and diffusion tensor imaging enabled the non-invasive reconstruction of white matter tracts through DW-MRI and DTI. Thereby making the construction of whole-brain level connectomes possible from the reconstructed microstructures.

The SC establishes a biological map upon which whole-brain dynamics function. It defines the network architecture that constrains neural communication, and thus, the possibility of functional interactions between certain regions. However, it does not capture how strongly these connections are functionally engaged or when they are active. Consequently, while the SC is crucial in WBM, providing the biological plausibility, it must be complemented by dynamic observables such as FC or COVτ to explain measured neural activity patterns [33].

Functional Connectivity
Functional connectivity refers to the statistical dependencies between neurophysiological signals recorded from different brain regions. It reflects the degree to which neural populations synchronize their activity over time, and it is typically inferred from temporal correlations in their measured signals. The most common approach involves computing the Pearson correlations between the time series of two regions. If two areas activate and deactivate simultaneously, they are said to be functionally connected. The Pearson correlation coefficient

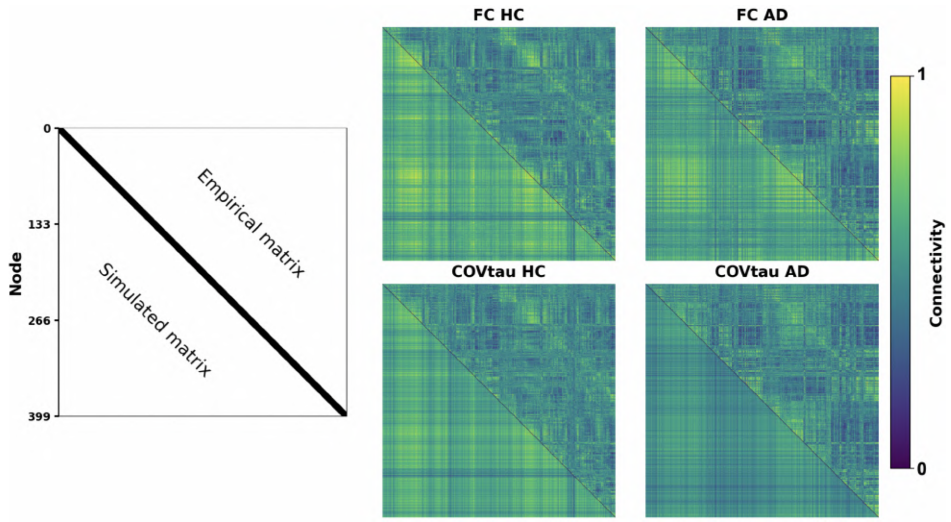

Figure 2.2.1: Functional connectivity and delayed covariance matrices from healthy and Alzheimer participants. The top right triangle shows the empirically determined values, the bottom left triangle shows the model-simulated values. The colorbar at the bottom of the figure shows the connectivity per pair of nodes.

between two time series X and Y of length N is defined as,

$$Corr(X,Y) = \frac{\sum_{n=1}^{N}(X_n - \bar{X})(Y_n - \bar{Y})}{\sqrt{\sum_{n=1}^{N}(X_n - \bar{X})^2}\sqrt{\sum_{n=1}^{N}(Y_n - \bar{Y})^2}}, \tag{2.2.1.1}$$

where $\bar{X}$ and $\bar{Y}$ denote the mean of the timeseries X and Y, respectively. In the top row of Fig. 2.2.1, we show two examples of FC matrices.

FC can be estimated from a range of modalities, including fMRI and E/MEG, each providing different spatial and temporal resolutions. By revealing robust large-scale networks (e.g., resting-state networks) in resting-state fMRI (rs-fMRI) [34], FC became a crucial observable in modern neuroscience. In contrast to task-based fMRI, resting-state fMRI measures spontaneous low-frequency (< 0.1 Hz) BOLD fluctuations in the absence of explicit stimulation. This type of fMRI is particularly convenient for clinical participants, including Alzheimer patients, as it does not require task ability and provides a direct window into large-scale network organization. Furthermore, the patterns in rs-fMRI-based FC have been shown to correlate with behavioral traits and disease states, making FC a preferred observable in clinical neuroscience [35].

Nevertheless, FC is limited in several respects. It does not account for the directionality or causality in interactions, is sensitive to signal normalization schemes, and typically captures only pairwise linear dependencies. Meanwhile, recent studies have emphasized the potential role of higher-order interactions, i.e., beyond pairwise correlations, in shaping large-scale neural dynamics [36], [37]. Therefore, although FC is essential for capturing select brain dynamics, it is recommended to complement one's toolbox with complementary dynamical observables.

Delayed Covariance Matrix
While classical FC is inherently dynamic, it exclusively quantifies instantaneous correlations between regions. Meanwhile, the dynamic FC (dFC) framework extends the instantaneous FC matrix by using a sliding window to capture changes in dynamics over the entire scan duration. This results in multiple FCs for a single scan.

Although, these matrices potentially encode a lot of dynamic information, they also significantly complicate the modeling. For that reason, we consider a more direct dynamic observable; the delayed-covariance matrix. The COVτ attempts to extend the dynamic features by incorporating temporal lags between neural signals. For two regions timeseries X and Y, the delayed covariance at time lag τ is defined as:

$$COV_{XY}(\tau) = \langle (X(t) - \bar{X})(Y(t+\tau) - \bar{Y}) \rangle_t, \tag{2.2.1.2}$$

where $\langle \cdot \rangle_t$ denotes temporal averaging. See the bottom row of Fig. 2.2.1 for two examples of COVτ matrices.

The exact value of the time lag τ is usually set to a time point at which the auto-correlation signal has decreased to about $1/e$. Practically, for BOLD time series with a repetition time (TR) of 2-3 seconds this means τ = TR. For further details, we refer to Appendix A.3.

Furthermore, unlike instantaneous FC, this measure captures how fluctuations in one region precede or follow those in another, providing information regarding directed or propagating dynamics within the network [38]. Thus, while dFC reveals that functional connectivity changes over time, COVτ provides mechanistic constraints on how the functional connectivity changes, potentially by imposing constraints on conduction velocity and coupling strength [39].

2.2.2. Networks: Effective Connectivity & Resting-State Networks

The increasing relevance of these connectivity observables in whole-brain modeling originates from the recognition that cognition arises from interactions between distributed neural regions rather than isolated brain areas. Whereas connectivity measures set the stage for describing dynamical patterns, network metrics describe how these patterns are intricately related. Together, they allow the construction of brain models that can reproduce empirical neural activity and, thus potentially bridge the gap between shape and function [7], [39], [40]. In the section, we outline two specific network modalities: the effective connectivity and the resting-state networks.

Effective Connectivity
Effective connectivity (EC) quantifies the directed, causal influence that one brain region exerts over another, aiming to capture the underlying mechanistic interactions that give rise to observed neural activity in a single matrix (Fig. 2.2.2). In contrast to the purely correlational nature of FC or COVτ, EC is not a direct empirical observable but a model-dependent construct that must be inferred from within a dynamical systems framework.

A widely used implementation of this construct is the generative effective connectivity (GEC) framework, as developed by Deco and colleagues [35], [41], in which the EC is defined as the set of directed coupling parameters of a stochastic whole-brain model whose noise-driven

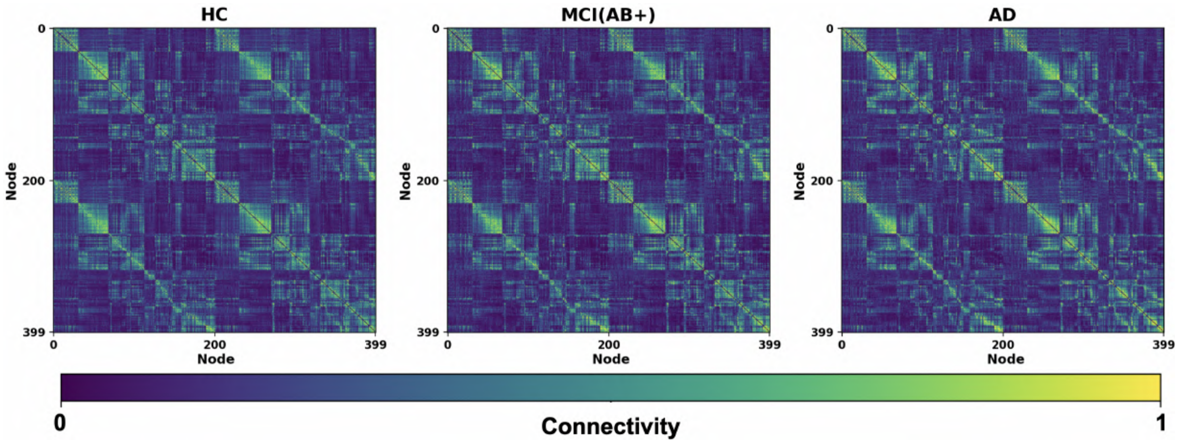

Figure 2.2.2: Average effective connectivity matrix per group as computed by the linear Hopf model used in this thesis. The colorbar at the bottom of the figure shows the connectivity per pair of nodes.

dynamics are fitted to reproduce empirical functional statistics, such as the FC and COVτ. In this formulation, EC reflects the effective causal interaction strengths shaped by structural connectivity, local dynamics, and noise. In Sec. 2.3 we explore the specific variant of GEC used in this thesis; the linearized Hopf/Stuart-Landau model.

Resting-State Networks

As mentioned, the success of resting-state fMRI studies originated through identifying large-scale brain networks: resting-state networks. In this part, we briefly discuss these networks and their relevance.

Traditionally, most fMRI studies have stuck with the stimulus or task paradigm. After recordings in which a certain stimulus or task occurred, the analysis primarily concerns itself with correlating the signal to the stimulus or task. However, some thirty years ago, Biswal and colleagues demonstrated that spontaneous BOLD fluctuations in the left and right motor cortex are synchronized, which suggests that there is a certain functional connectivity at rest [42]. This gave rise to a new direction of research; resting-state fMRI and resting-state networks. Generally, these networks show low frequency (<0.1 Hz) patterns of spontaneous, correlated brain activity while the subject is not exposed to any stimulus or performing any task.

The main arguments for rs-fMRI over traditional stimulus-based fMRI approaches are increased applicability and improved reproducibility. In traditional fMRI, the different types of stimuli and the subjectiveness of task performance result in additional variability that hinders comparisons across studies. Furthermore, some groups, such as infants, elderly, and non-human animals, are at times difficult to include in the studies, limiting the generality of the method [27]. Notably, AD patients are often unable to perform certain tasks or reliably respond to stimuli. In this case, focusing on RSN analysis increases robustness of results and extends the scope of possible studies.

A major step in the characterization of RSNs was made by Yeo and colleagues in 2011, who applied large-scale data-driven clustering on resting-state fMRI data from 1,000 healthy individuals to derive a cortical parcellation of seven large-scale networks [34]. As shown

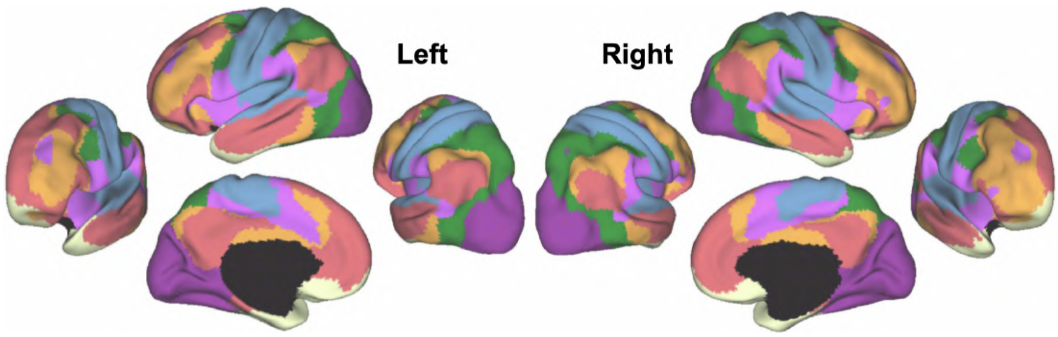

Figure 2.2.3: The 7 resting state networks of the human cerebral cortex from [34] based on 1,000 subjects. Views from both hemisphere are shown and the following networks are colored: visual, somatomotor, dorsal attention, ventral attention, limbic, frontopariental control, and default mode. The regions in black are subcortical areas.

in Fig. 2.2.3, these include the visual, somatomotor, dorsal attention, ventral attention, limbic, frontoparietal control, and default mode networks. Importantly, the parcellation underlying these networks has become widely adopted for studying functional organization and network-level alterations in neurological and psychiatric conditions [41], [43].

Recent efforts provide some evidence for connecting task/stimulus activation to RSNs through the active cortex model [44]. This model postulates that all task dynamics arise through the selective activation and modulation of RSNs, suggesting that the resting state represents the fundamental basis from which all cognitive activations emerge. In this view, task-evoked activity does not activate new functional networks but rather reconfigures the baseline activity of existing networks. As such, the framework unifies resting and task paradigms, providing a continuous connection between spontaneous and perturbed dynamics. In the neurodegenerative context, this perspective is particularly valuable: it implies that pathological changes in connectivity patterns, as seen in AD, can directly constrain or distort task-related activations.

2.3. Stuart-Landau Oscillators and its Linear Approximation

To conclude our introduction to the field of WBM, we discuss the actual modeling part by first describing the local node dynamics. Although there exist many different node dynamics, we limit this description to the phenomenological brain model used in this study, the Stuart-Landau/Hopf model and in particular its linear approximation.

2.3.1. Stuart-Landau Oscillators

First introduced by Matthews and Strogatz in 1990 [45], this canonical model is developed to study systems of coupled nonlinear oscillators for which both the phase and amplitude interact. It is known as the Stuart-Landau model or, in the neuroscience community, as the Hopf model [7].

The local dynamics of a node are described by the normal form of a supercritical Hopf bifurcation:

$$\frac{dz}{dt} = \dot{z} = (a + i\omega)z - |z|^2 z + \eta, \tag{2.3.1.1}$$

where $z = x + iy$, with x and y are the real and imaginary parts of the state variable z (arbitrary units), the module of z is given by $|z|$, $\omega = 2\pi\nu$ is the intrinsic angular frequency ($rads^{-1}$) with ν the intrinsic frequency in Hz, and a is the bifurcation parameter in s^{-1}. Additive white noise is denoted by η, i.e. $\langle\eta(t)\rangle = 0$ and $\langle\eta(t)\eta(t')\rangle = \sigma^2\delta(t - t')$, with σ the noise amplitude and the angular brackets $\langle.\rangle$ represents the average over stochastic realizations. In whole-brain models based on fMRI data, the real part of the state variable, $x = Re(z)$, denotes the simulated BOLD signals and the intrinsic frequencies are estimated by taking the top signal frequency of per node.

To describe the dynamics, we can write Eq. 2.3.1.1 in polar coordinates without noise, i.e. $z = re^{i\theta}$,

$$\frac{d\theta}{dt} = \omega \tag{2.3.1.2}$$

$$\frac{dr}{dt} = ar - r^3 \tag{2.3.1.3}$$

where $\theta = \arctan(y/x)$ is the phase and $r = |z|$ the module of z. Now, we see in Eq. 2.3.1.3 that a fixed point is given by $r = 0$ and the stability of this point depends on the bifurcation parameter a. The eigenvalues of the system are $\lambda = a \pm i\omega$. For $a < 0$, both eigenvalues have a negative real part, and the point $r = 0$ is stable as fluctuations around this solution are damped (i.e., the spiral solution). Note that when we add noise, oscillations are induced. For $a > 0$, $r = 0$ is unstable and fluctuations are amplified. Also, a new fixed point appears at $r = a^{1/2}$. This solution gives rise to a limit cycle (Fig. 2.3.1).

In the case of N nodes, such as in a whole-brain model, the nodes i and j are coupled through the component C_{ij} of connectivity matrix $\mathbf{C}$. The dynamics are then given by,

$$\frac{dz_i}{dt} = [(a_i + i\omega_i) - |z_i|^2]z_i + g\sum_{j=1}^{N} C_{ij}(z_j - z_i) + \eta_i, \tag{2.3.1.4}$$

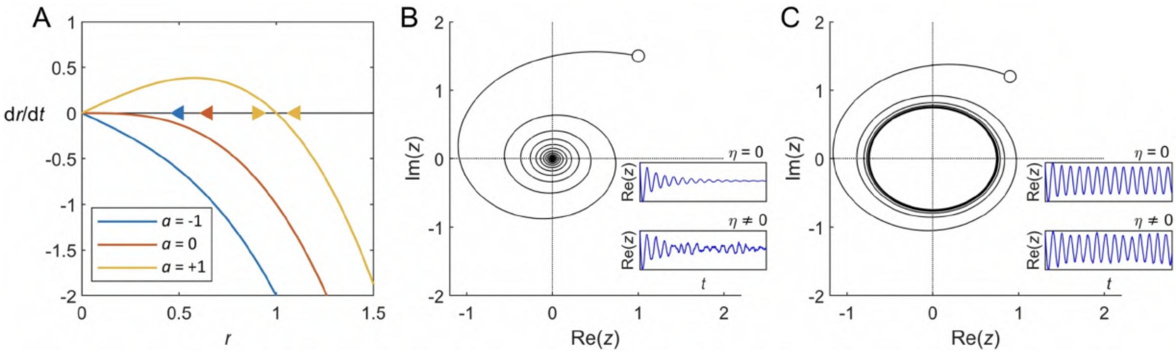

Figure 2.3.1: Stuart-Landau model dynamics. (**A**) The fixed points of a node have modules given by the roots of $\dot{r} = ar - r^3$. For $a < 0$, the solution $r = 0$ is stable since fluctuations around it are decreased (i.e., $\dot{r} < 0$). However, when $a > 0$, $r = 0$ is unstable as deviations from it are amplified (i.e., $\dot{r} > 0$). In this second case, another fixed point appears that is given by $r = a^{1/2}$, which is stable since fluctuations around it, $r = a^{1/2} + \delta r$, are increased if $\delta r < 0$, but decreased if $\delta r > 0$. The arrows indicate the direction of flow and are given by the sign of $\dot{r}$. (**B**) Single-node behavior for $a < 0$. The system relaxes into a spiral solution from the initial condition (white circle) to the origin of the complex plane. (**C**) Single-node behavior for $a > 0$. The system drives itself through oscillations (limit cycle). In both **B** and **C**, the insets show the node dynamics in absence (top) and presence (bottom) of noise η. Figure adapted from [8].

where g denotes the global coupling parameter. This parameter corresponds to a scaling factor for all synaptic conductivity connections. Note that the connectivity matrix can be used, which is often done, as fitting parameter, thereby yielding the effective connectivity as outcome. In these instances, the structural connectivity is commonly used for initialization.

2.3.2. Linearization

Computing the network measures and statistics of Eq. 2.3.1.4 requires long simulation times, which hinders parameter exploration withing the fitting process. Although the Hopf model contains nonlinear oscillators, previous studies have used resting-state neuroimaging data to suggest that nonlinearities are small. This allows for the linearization and analytical estimation of the model's statistics, thereby greatly increasing its practicality in simulating large quantities of data [8].

In vector form, Eq. 2.3.1.4 can be expressed as,

$$\frac{d\mathbf{z}}{dt} = (\mathbf{a} - g\mathbf{S} + i\omega) \odot \mathbf{z} - (\mathbf{z} \odot \overline{\mathbf{z}})\mathbf{z} + g\mathbf{C}\mathbf{z} + \eta, \tag{2.3.2.1}$$

where bold symbols are used to indicate vectors and matrices (i.e. $\mathbf{z} = [z_1, ..., z_N]^T$), $\overline{\mathbf{z}}$ is the complex conjugate of $\mathbf{z}$, $\mathbf{S}$ is the vector with elements $S_i = \sum_j C_{ij}$, and the $\odot$ operator denotes the Hadamard element-wise product.

In the linearized version, fluctuations $\delta\mathbf{z}$ are evaluated around the stationary point $\mathbf{z} = 0$. Higher-order terms, i.e., $(\delta\mathbf{z} \odot \delta\overline{\mathbf{z}})\delta\mathbf{z}$, are discarded and only first order in $\delta\mathbf{z}$ terms are kept. As a result, the system is reduced to a Langevin equation,

$$\frac{d}{dt}\delta\mathbf{u} = \mathbf{A}\delta\mathbf{u} + \eta, \tag{2.3.2.2}$$

where $\delta\mathbf{u} = (\delta\mathbf{x}, \delta\mathbf{y})$ is the $2N$-dimensional vector containing the fluctuations of real and imaginary parts and $\mathbf{A}$ is the $2N \times 2N$ Jacobian matrix of the system evaluated at the fixed point:

$$A_{jk} = \frac{\partial F_j}{\partial u_k}\bigg|_0, \tag{2.3.2.3}$$

where $F_j = (a_j - x_j^2 - y_j^2)x_j - \omega_j y_j + g\sum_{k=1}^N C_{jk}(x_k - x_j)$ for $1 \leq j \leq N$ (real parts), and $F_j = (a_j - x_j^2 - y_j^2)y_j - \omega_j x_j + g\sum_{k=1}^N C_{jk}(y_k - y_j)$ for $N+1 \leq j \leq 2N$ (imaginary parts).

When we evaluate the partial derivatives at the fixed point, we can write $\mathbf{A}$ as a block matrix,

$$\mathbf{A} = \begin{bmatrix} \mathbf{A_{xx}} & \mathbf{A_{xy}} \\ \mathbf{A_{yx}} & \mathbf{A_{yy}} \end{bmatrix}, \tag{2.3.2.4}$$

with $\mathbf{A_{xx}} = \mathbf{A_{yy}} = \mathrm{diag}(\mathbf{a} - g\mathbf{S}) + g\mathbf{C}$, and $\mathbf{A_{xy}} = -\mathbf{A_{yx}} = \mathrm{diag}(\omega)$. Now, we note that the Jacobian matrix determines the statistics of the linear system [8].

Using $\mathbf{A}$, we can compute the covariance matrix $\mathbf{C_v}$ by solving the continuous-time Lyapunov equation under the assumption of stationarity, i.e., $\frac{d}{dt}\delta\mathbf{u} = 0$,

$$\mathbf{A}\mathbf{C_v} + \mathbf{C_v}\mathbf{A^T} + \mathbf{Q} = 0, \tag{2.3.2.5}$$

where matrix $\mathbf{Q} = \mathrm{diag}_{2N\times 2N}(\sigma_\eta^2)$ encodes the additive noise. The covariance matrix $\mathbf{C_v}$ allows us then to calculate the connectivity matrices. In particular, the functional connectivity (through Eq. 2.2.1.1) and the τ-delayed covariance matrix. In continuous-time dynamical systems, the relationship between two variables at different times is given by the state propagator. Specifically, the COVτ is determined by,

$$\mathbf{C_v}(t+\tau) = \mathbf{C_v} \cdot \exp(\tau \cdot \mathbf{A}), \tag{2.3.2.6}$$

where the exponential is determined by the scaling and squaring method [46]. Furthermore, both the empirical and the simulated covariance matrices are normalized by $\sqrt{\mathbf{C_v}_{ii}\mathbf{C_v}_{jj}}$ for each pair (i, j).

After computing these observables, we construct the effective connectivity matrix, $\mathbf{C}$, by iteratively minimizing the multi-objective cost function,

$$\begin{aligned} L_{EC} = &\ \mathrm{MSE}(\mathbf{FC}_{emp}, \mathbf{FC}_{sim}) + \mathrm{MSE}(\mathbf{COV}\tau_{emp}, \mathbf{COV}\tau_{sim}) \\ &+ \left[1 - \mathrm{Corr}(\mathbf{FC}_{emp}, \mathbf{FC}_{sim})\right] + \left[1 - \mathrm{Corr}(\mathbf{COV}\tau_{emp}, \mathbf{COV}\tau_{sim})\right], \end{aligned} \tag{2.3.2.7}$$

where MSE denotes the mean squared error and $Corr$ the Pearson correlation as formulated in Eq. 2.2.1.1. Whereas occasionally the global fitting parameter is included, the main degrees of freedom of the model are given by the EC matrix. However, in this study, the fitting metrics are expanded to include individual node bifurcation and noise parameters. For the exact modeling procedure and further fit parameters, we refer to Appendix A.3.

3

The Fluctuation-Dissipation Theorem

When a system is externally disturbed and driven away from equilibrium, it naturally evolves back toward equilibrium through an entropy-producing process; this is what we call dissipation. Conversely, when the system is left undisturbed for a period of time, it may on occasions spontaneously deviate from equilibrium; we refer to this as fluctuation. Although appearing to belong to different domains; dissipation is induced and fluctuation arises spontaneously, it turns out that these physical phenomena are related through a relationship described by the fluctuation-dissipation theorem (FDT). Notably, this relationship is exclusively valid in detailed balance equilibrium. Consequently, the system's distance from equilibrium can be quantified by measuring the degree to which it violates the FDT.

Originally formulated for near-equilibrium systems by Callen and Welton [4] and Kubo [5], the FDT formulates a quantitative relationship between spontaneous fluctuations and linear response to small perturbations. Even though the brain operates far from detailed balance, i.e. a condition in which each elementary process is in equilibrium with its reverse process, generalized formulations of the FDT have extended the framework to out of equilibrium systems [47]. Thereby enabling the quantification of deviations from detailed balance equilibrium, specifically in Langevin systems [48]. Such deviations provide a measure of non-equilibrium and thereby hierarchical organization and directional information flow, metrics not easily accessible through static measures.

In this chapter, we start in Sec. 3.1 by presenting a Langevin formulation of the FDT and describing the equilibrium violation metrics. Next, in Sec. 3.2, we present a model-free implementation of the FDT framework. Finally, in Sec. 3.3, we introduce a model-based analytical implementation.

3.1. Basic Principles

Let us begin by considering a system described by a Hamiltonian H_0, quenched from a heat bath of temperature T at time $t = 0$. The auto-correlation function $C_O(t, t')$ of a local observable O at subsequent times t and t' is given by,

$$C_O(t, t') = \langle O(t)O(t') \rangle, \tag{3.1.0.1}$$

where $\langle . \rangle$ denote the ensemble averages. In detailed balance equilibrium, for a given $t - t'$ and large t, the auto-correlation function is time-translation invariant (TTI), i.e., $C_O(t, t') = C_O(t - t')$. Note that this property of homogeneity in time is a strong indicator of detailed balance in this context. When we introduce a small enough perturbation $h(t')$ at time t' to the system, the variation of the observable $\langle O(t) \rangle$ due to this perturbation is given by the response function. Specifically, given the perturbed Hamiltonian,

$$H = H_0 + \int h(t')O(t')dt', \tag{3.1.0.2}$$

we consider the following linear response function,

$$R(t, t') = \frac{\delta \langle O(t) \rangle}{\delta h(t')}, \tag{3.1.0.3}$$

where we note that this gives zero if $t' > t$, due to causality. The response function thus relates to the perturbation-induced change in the observable in a linear fashion,

$$O(t) = \int_{t'}^{t} R(t, t')h(t')dt'. \tag{3.1.0.4}$$

In equilibrium, the response function is related to the auto-correlation function through the fluctuation-dissipation theorem,

$$R_{eq}(t, t') = \beta\theta(t - t') \frac{\partial C_O(t, t')}{\partial t'}, \tag{3.1.0.5}$$

where $\beta = 1/T$ and $\theta(t - t')$ captures the causality condition $t \geq t'$ [48]. Moreover, note that the relation is TTI as well, i.e., $R(t, t') = R(t - t')$. However, outside of detailed balance equilibrium the time-translation invariance does not hold. In this regime, the exact FDT (3.1.0.5) is broken and we have to consider the generalized relation between fluctuation and dissipation,

$$R_{non-eq}(t, t') = X(t, t')R_{eq}(t, t'). \tag{3.1.0.6}$$

Here, the fluctuation-dissipation ratio $X(t, t')$, a function of both t and t', gives a measure of equilibrium violation through quantifying the difference between the in and out of equilibrium response functions.

3.1.1. Langevin Equation

There are many different formulations and derivations of the FDT. In general, the literature makes a distinction between two families of approaches: deterministic and stochastic. Under the deterministic umbrella fall derivations using linear response theory [49], operator algebra [50], and quantum statistical physics [51]. Stochastic approaches focus on the Focker-Planck and Langevin dynamics [47]. Considering the scope of this thesis, only we present a brief overview of the Langevin approach.

Originally, the Langevin equation was introduced to describe Brownian motion. However, in general, this stochastic differential equation describes how a system evolves under the combined influence of deterministic and fluctuating forces. We consider the following instance: a system described by variable $y(t)$ is subjected to the driving force $F_y(t)$ and Gaussian random noise $\eta(t)$,

$$\frac{dy(t)}{dt} = -F_y(t) + \eta(t), \tag{3.1.1.1}$$

where the Gaussian random noise has a zero mean and satisfies the following relation,

$$\langle \eta(t)\eta(t')\rangle = 2T\delta(t-t'), \tag{3.1.1.2}$$

with $T = \frac{\sigma^2}{2}$ and σ the standard deviation of $\eta(t)$. Taking the auto-correlation difference of the variable y and using Eq. 3.1.1.1, we find,

$$\begin{aligned}
(\frac{\partial}{\partial t'} - \frac{\partial}{\partial t})C_y(t,t') &= \langle y(t)\frac{\partial y(t')}{\partial t'}\rangle - \langle \frac{\partial y(t)}{\partial t} y(t')\rangle && (3.1.1.3)\\
&= -\langle y(t)[F_y(t') - \eta(t')]\rangle + \langle [F_y(t) - \eta(t)]y(t')\rangle && (3.1.1.4)\\
&= \langle F_y(t)y(t')\rangle - \langle y(t)F_y(t')\rangle + \langle y(t)\eta(t')\rangle - \langle \eta(t)y(t')\rangle && (3.1.1.5)\\
&= A(t,t') + \langle y(t)\eta(t')\rangle, && (3.1.1.6)
\end{aligned}$$

where in the last equality we define $A(t,t') \equiv \langle F_y(t)y(t')\rangle - \langle y(t)F_y(t')\rangle$ and imply $t \geq t'$ causality, i.e., $\langle \eta(t)y(t')\rangle = 0$.

Given that our variable is a noise-dependent functional, i.e., it satisfies $y(t) = G[\eta](t)$ with the exact relation specified in Eq. 3.1.1.1, we can use Novikov's theorem [52]. This result states that for Gaussian white noise and any stochastic functional $y[\eta](t)$ we have,

$$\begin{aligned}
\langle y(t)\eta(t')\rangle &= \int \langle \eta(t')\eta(s)\rangle \left\langle \frac{\delta y(t)}{\delta \eta(s)} \right\rangle ds && (3.1.1.7)\\
&= \int 2T\delta(s-t')R(t,s)ds && (3.1.1.8)\\
&= 2TR(t,t'), && (3.1.1.9)
\end{aligned}$$

where we used Eq. 3.1.1.2 and the fact that Gaussian noise takes the role of a perturbation in Eq. 3.1.1.8.

Now, considering that in detailed balance equilibrium, time-reversal symmetry (TRS) dictates that $\langle B(t)D(t')\rangle = \langle B(t')D(t)\rangle$ for B and D both functions of $y(t)$, we find that $A(t,t') = 0$. Furthermore, note that the derivative of the auto-correlation is TTI in equilibrium, i.e., $\frac{\partial C(t,t')}{\partial t'} = -\frac{\partial C(t,t')}{\partial t}$, and we recover the equilibrium FDT,

$$R_{eq}(t,t') = \beta \frac{\partial C_y(t,t')}{\partial t'}, \tag{3.1.1.10}$$

for $t \geq t'$.

However, more interesting for our outside of equilibrium considerations, we find a more precise expression of the FDT violation,

$$R(t,t') = \frac{\beta}{2}\left[\frac{\partial C_y(t,t')}{\partial t'} - \frac{\partial C_y(t,t')}{\partial t} - A(t,t')\right], \tag{3.1.1.11}$$

which reveals a formulation of $X(t,t')$ as denoted in Eq. 3.1.0.6,

$$X(t,t') = \frac{1}{2}\left[1 - \frac{A(t,t')}{\partial C_y(t,t')/\partial t'} - \frac{\partial C_y(t,t')/\partial t}{\partial C_y(t,t')/\partial t'}\right]. \tag{3.1.1.12}$$

The two non-trivial contributions that determine the fluctuation–dissipation ratio in (3.1.1.12) originate from violations of time-reversal symmetry (TRS) and time-translation invariance (TTI), respectively. A violation of TRS indicates the presence of non-conservative probability currents and implies net entropy production, corresponding to a breakdown of detailed balance. By contrast, a violation of TTI signals the absence of a steady-state, as stationary systems are characterized by time-independent macroscopic statistics despite ongoing microscopic dynamics. Non-stationarity may arise either from external driving or from intrinsic aging processes, in which case TRS may remain intact.

Since we explicitly model the dynamics using a Langevin equation with potentially non-conservative drift, violations of detailed balance are expected. Importantly, FDT-based violation metrics are most interpretable in non-equilibrium steady states, where TTI holds and deviations from the equilibrium FDT isolate irreversibility and entropy production alone.

3.1.2. FDT Violations

In systems satisfying detailed balance, the fluctuation–dissipation theorem establishes a direct relation between the linear response function $R(t,t')$ and the spontaneous fluctuations encoded in the auto-correlation function $C(t,t')$. As derived in Sec. 3.1.1, departures from equilibrium introduce two distinct sources of FDT violation: the breaking of time-reversal symmetry, reflected in the antisymmetric correlator $A(t,t')$, and the breakdown of time-translation invariance, encoded in the asymmetry between the temporal derivatives of $C(t,t')$.

A convenient differential measure of the FDT violation was introduced by [48] as

$$V(t,t') \equiv \frac{\partial C(t,t')}{\partial t'} - TR(t,t') = [1 - X(t,t')]\frac{\partial C(t,t')}{\partial t'}, \tag{3.1.2.1}$$

where $X(t,t')$ denotes the fluctuation-dissipation ratio derived in Eq. 3.1.1.12. In equilibrium, where both TRS and TTI hold, one has $X(t,t') = 1$ and therefore $V(t,t') = 0$. Any deviation from zero thus constitutes a direct signature of non-equilibrium dynamics.

Since numerical differentiation is sensitive to noise amplification, we focus on the corresponding integrated violation (IV),

$$I(t,t') \equiv \int_{t'}^{t} V(t,s)\,ds \tag{3.1.2.2}$$

$$= C(t,t) - C(t,t') - T\chi(t,t'), \tag{3.1.2.3}$$

where

$$\chi(t,t') \equiv \int_{t'}^{t} R(t,s)\,ds \tag{3.1.2.4}$$

is the integrated response (dynamic susceptibility). The quantity $I(t,t')$ measures the cumulative mismatch between spontaneous fluctuations and linear response and therefore provides a robust scalar measure of the deviation from detailed balance.

To be able to compare the distance-from-equilibrium metrics across different modalities, we use a non-equilibrium steady-state variance-normalized description of the integral violation. First, to ensure non-equilibrium steady-state dynamics, i.e., statistics that are independent of the initial conditions, we take the time lag $\tau = t - t'$ sufficiently large. Second, to ensure comparability across modalities, we require a mapping to a dimensionless metric that is invariant under rescalings $u_i \mapsto \alpha_i u_i$. This is commonly achieved through z-scoring,

$$u_i(t) \mapsto \frac{u_i(t) - \mu_i}{\sigma_i}, \tag{3.1.2.5}$$

where μ_i and σ_i denote the temporal mean and standard deviation of the time series $u_i(t)$ of parcel i.

While centering operations such as $\mathbf{u}(t) - \boldsymbol{\mu}$ act uniformly on vectors and bilinear forms, linear rescalings do not. Specifically, for a linear transformation $\boldsymbol{u} = \mathbf{S}\boldsymbol{z}$ one finds,

$$\text{vectors:} \quad \boldsymbol{z} = \mathbf{S}^{-1}\boldsymbol{u}, \tag{3.1.2.6}$$

$$\text{bilinear forms:} \quad \tilde{\mathbf{B}} = \mathbf{S}^{-1}\mathbf{B}\,\mathbf{S}^{-1}. \tag{3.1.2.7}$$

We therefore consider the non-equilibrium steady-state covariance matrix,

$$\mathcal{V}_\infty \equiv \lim_{t\to\infty} \mathbf{C}_u(t,t) = \lim_{t\to\infty} \langle \mathbf{u}(t)\mathbf{u}(t) \rangle, \tag{3.1.2.8}$$

and introduce the diagonal rescaling matrix,

$$\mathbf{S} \equiv \boldsymbol{\Delta}_\infty^{-1/2}, \quad \text{where} \quad \boldsymbol{\Delta}_\infty \equiv \mathrm{Diag}(\mathcal{V}_\infty). \tag{3.1.2.9}$$

The corresponding linear variable transformation,

$$\mathbf{z}(t) = \mathbf{S}^{-1}\mathbf{u}(t) \tag{3.1.2.10}$$

implements the division step of the z-scoring procedure. For bilinear forms such as the covariance, the associated transformation becomes

$$\overline{\mathcal{V}}_\infty = \mathbf{S}^{-1}\mathcal{V}_\infty\mathbf{S}^{-1}. \tag{3.1.2.11}$$

This distinction between vector and bilinear transformations is crucial for the correct normalization of the integral violation metric and are applied in both the model-free and analytical implementations.

3.2. Model-free Implementation

To find an estimation of the FDT violation as outlined above, no model is required. For many systems, the fluctuation-dissipation relation can be directly approximated from empirical observables, in particular from timeseries such as the BOLD signal. After assuming Langevin dynamics, the only requirements are the existence of a $\langle y(t)\eta(t')\rangle$ relation and the ability to approximate it. The initial implementation of this model-free framework was first constructed in [9] where it was also used in an Alzheimer context. Here, we briefly introduce a version of that implementation.

Consider the variable $y(t)$ to directly correspond to the measured BOLD signal. As mentioned, we assume the dynamics to be governed by the Langevin equation, Eq. 3.1.1.1. That means we are able to use the previously constructed framework to determine the integral violation, as specified by Eq. 3.1.2.2. To start, we calculate the instantaneous derivative, $\dot{y} = \frac{dy(t)}{dt}$, of the signal. Next, by applying a Wiener filter, we determine the deterministic and the stochastic part, $-F_y(t)$ and $\eta(t)$, respectively (Fig. 3.2.1**A**). The Wiener filter is an optimal linear filter that minimizes the mean squared error between the estimated signal and the desired signal, given some known properties of the signal and noise. In this case, we assume that the noise $\eta(t)$ is the average of the local variance of the input and that it satisfies the Gaussian relation of Eq. 3.1.1.2. We can easily verify that this condition is satisfied (Fig. 3.2.1**C**). After splitting

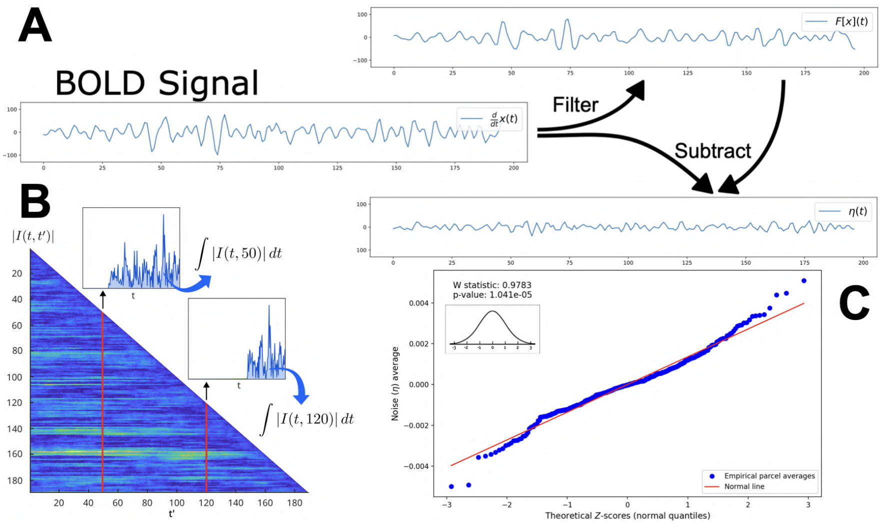

Figure 3.2.1: Overview of the model-free IV-FDT implementation. Using the Wiener filter, the BOLD signal is split into a deterministic and a stochastic part (**A**) [9]. The stochastic noise part is checked for Gaussian properties (**C**). After splitting the signal, we use Novikov's theorem to compute the integral violation of the detailed balance equilibrium. To ensure that the non-equilibrium steady-state version is calculated, causality is enforced and a time lag condition is introduced (**B**) [16].

the signal and enforcing causality, i.e. $t \geq t'$, we are able to estimate the integral violation by making use of the connection between $\langle y(t)\eta(t')\rangle$ and the linear response in Eq. 3.1.1.8. Moreover, to approximate the non-equilibrium steady-state version of the integral violation, we introduce a time lag condition $\tau = t - t' \geq 20$. Finally, we obtain our variance-normalized empirical estimation of the integral violation by temporal averaging over subsequent time lags and normalizing the measure with respect to the steady-state covariance.

3.3. Analytical Implementation

The first Stuart-Landau model-based formulation of the FDT framework was introduced by Deco et al. [15] and augmented by Patow et al. [9] and Monti et al. [16]. These works highlighted the FDT framework's potential for identifying out-of-equilibrium dynamics in, respectively, cognitive tasks, Alzheimer's disease, and wakefulness/sleep, using the non-linear Stuart-Landau brain model. In this section, we extend this model-based implementation to an analytical one.

The linearized Stuart-Landau model and the Langevin FDT introduced in Sec. 3.1.1 are closely related. Recall that in Sec. 2.3.2 we found that the linearization of the Stuart-Landau model led to a Langevin equation, Eq. 2.3.2.2. This equation is mathematically identical to Eq. 3.1.1.1 and can be solved exactly. This allows us to find analytical expressions of the Langevin FDT in terms of model-determined parameters. In particular, we use the model-determined Jacobian matrix $\mathbf{A}$, Eq. 2.3.2.4, to exactly compute the integral violation metric. In this section, we present this analytical framework as derived by J.Monti (*J.Monti, personal communication, Dec. 2025*).

Starting from an N-dimensional version of Eq. 2.3.2.2,

$$\frac{d\boldsymbol{u}(t)}{dt} \;=\; -\,\boldsymbol{\Gamma}\,\boldsymbol{u}(t) \;+\; \boldsymbol{\eta}(t), \tag{3.3.0.1}$$

where $\boldsymbol{\Gamma} = -\mathbf{A} \in \mathbb{R}^{N\times N}$ is constant in time and $\boldsymbol{\eta}(t)$ is a zero-mean Gaussian noise with covariance

$$\langle \boldsymbol{\eta}(t)\,\boldsymbol{\eta}(t')^\top \rangle = \mathbf{Q}\,\delta(t-t'), \tag{3.3.0.2}$$

a slightly more general version of Eq. 3.1.1.2, where $\mathbf{Q}$ is the noise covariance-matrix. We denote the initial condition $\boldsymbol{u}(0) = \boldsymbol{x}_0$ with $\langle \boldsymbol{u}_0 \rangle = \mathbf{0}$, and $\langle \boldsymbol{u}_0\,\boldsymbol{u}_0^\top \rangle = \boldsymbol{\mathcal{V}}_0$ is the initial condition covariance matrix. Furthermore, we assume $t \geq t'$ unless otherwise stated.

Using Duhamel's formula, we find the unique solution to the N-dimensional Langevin equation (Appendix A.5),

$$\boldsymbol{u}(t) \;=\; e^{-\boldsymbol{\Gamma} t}\,\boldsymbol{u}_0 \;+\; \int_0^t e^{-\boldsymbol{\Gamma}(t-\tau)}\,\boldsymbol{\eta}(\tau)\,d\tau. \tag{3.3.0.3}$$

Now recall that to compute the integral violation of Eq. 3.1.2.3, we need to find expressions for the auto-correlation function and the linear response function.

Auto-correlation function

Let us start by considering the analytic expression of the auto-correlation function of Eq. 3.1.0.1. Inserting our unique solution (3.3.0.3), taking the expectation, and using that $\langle \boldsymbol{u}_0\,\boldsymbol{\eta}(\cdot)^\top \rangle = \mathbf{0}$

and $\langle \boldsymbol{\eta}(\tau)\,\boldsymbol{\eta}(\sigma)^\top\rangle = \mathbf{Q}\,\delta(\tau-\sigma)$, we find,

$$\mathbf{C}_u(t,t') = \left\langle \boldsymbol{u}(t)\,\boldsymbol{u}(t')^\top \right\rangle \tag{3.3.0.4}$$

$$= e^{-\boldsymbol{\Gamma} t}\,\boldsymbol{\mathcal{V}}_0\,e^{-\boldsymbol{\Gamma}^\top t'} \;+\; \int_0^t \int_0^{t'} e^{-\boldsymbol{\Gamma}(t-\tau)}\,\langle \boldsymbol{\eta}(\tau)\,\boldsymbol{\eta}(\sigma)^\top\rangle\,e^{-\boldsymbol{\Gamma}^\top(t'-\sigma)}\,d\sigma\,d\tau$$

$$= e^{-\boldsymbol{\Gamma} t}\,\boldsymbol{\mathcal{V}}_0\,e^{-\boldsymbol{\Gamma}^\top t'} \;+\; \int_0^{t'} e^{-\boldsymbol{\Gamma}(t-\tau)}\,\mathbf{Q}\,e^{-\boldsymbol{\Gamma}^\top(t'-\tau)}\,d\tau, \qquad (t \geq t'). \tag{3.3.0.5}$$

Where the double integral collapsed to a single one because of the Dirac delta $\delta(\tau-\sigma)$, and the upper limit is t' (not t) by causality when $t \geq t'$. Considering the integral term in (3.3.0.5), we rewrite,

$$\int_0^{t'} e^{-\boldsymbol{\Gamma}(t-\tau)}\mathbf{Q}e^{-\boldsymbol{\Gamma}^\top(t'-\tau)}d\tau = e^{-\boldsymbol{\Gamma} t}\left[\int_0^{t'} e^{\boldsymbol{\Gamma}\tau}\mathbf{Q}e^{\boldsymbol{\Gamma}^\top\tau}d\tau\right]e^{-\boldsymbol{\Gamma}^\top t'} \tag{3.3.0.6}$$

Now regard the non-equilibrium steady-state covariance matrix $\boldsymbol{\mathcal{V}}_\infty$, Eq. 3.1.2.8, as solution to the continuous-time Lyapunov equation,

$$\boldsymbol{\Gamma}\boldsymbol{\mathcal{V}}_\infty + \boldsymbol{\mathcal{V}}_\infty\boldsymbol{\Gamma}^\top = \mathbf{Q}, \tag{3.3.0.7}$$

this solution is unique if all eigenvalues of $\boldsymbol{\Gamma}$ have positive real parts. Now, given that the τ-derivative of $e^{\boldsymbol{\Gamma}\tau}\boldsymbol{\mathcal{V}}_\infty e^{\boldsymbol{\Gamma}^\top\tau}$ is,

$$\frac{d}{d\tau}\left[e^{\boldsymbol{\Gamma}\tau}\boldsymbol{\mathcal{V}}_\infty e^{\boldsymbol{\Gamma}^\top\tau}\right] = \boldsymbol{\Gamma}e^{\boldsymbol{\Gamma}\tau}\boldsymbol{\mathcal{V}}_\infty e^{\boldsymbol{\Gamma}^\top\tau} + e^{\boldsymbol{\Gamma}\tau}\boldsymbol{\mathcal{V}}_\infty\boldsymbol{\Gamma}^\top e^{\boldsymbol{\Gamma}^\top\tau} \tag{3.3.0.8}$$

$$= e^{\boldsymbol{\Gamma}\tau}\left(\boldsymbol{\Gamma}\boldsymbol{\mathcal{V}}_\infty + \boldsymbol{\mathcal{V}}_\infty\boldsymbol{\Gamma}^\top\right)e^{\boldsymbol{\Gamma}^\top\tau} \tag{3.3.0.9}$$

$$= e^{\boldsymbol{\Gamma}\tau}\mathbf{Q}e^{\boldsymbol{\Gamma}^\top\tau}, \tag{3.3.0.10}$$

the fundamental theorem of calculus returns to us that,

$$\int_0^s e^{\boldsymbol{\Gamma}\tau}\mathbf{Q}e^{\boldsymbol{\Gamma}^\top\tau}d\tau = \left[e^{\boldsymbol{\Gamma}\tau}\boldsymbol{\mathcal{V}}_\infty e^{\boldsymbol{\Gamma}^\top\tau}\right]_0^s \tag{3.3.0.11}$$

$$= e^{\boldsymbol{\Gamma}s}\boldsymbol{\mathcal{V}}_\infty e^{\boldsymbol{\Gamma}^\top s} - \boldsymbol{\mathcal{V}}_\infty. \tag{3.3.0.12}$$

Inserting this expression into (3.3.0.5) gives us the closed form of the auto-correlation function,

$$\mathbf{C}_u(t,t') = e^{-\boldsymbol{\Gamma}(t-t')}\boldsymbol{\mathcal{V}}_\infty + e^{-\boldsymbol{\Gamma} t}\left(\boldsymbol{\mathcal{V}}_0 - \boldsymbol{\mathcal{V}}_\infty\right)e^{-\boldsymbol{\Gamma}^\top t'}. \tag{3.3.0.13}$$

Now we can see that,

$$\mathbf{C}_u(t,t) = \boldsymbol{\mathcal{V}}_\infty + e^{-\boldsymbol{\Gamma} t}\left(\boldsymbol{\mathcal{V}}_0 - \boldsymbol{\mathcal{V}}_\infty\right)e^{-\boldsymbol{\Gamma}^\top t}, \tag{3.3.0.14}$$

and that,

$$\mathbf{C}_u(t,t') = e^{-\boldsymbol{\Gamma}(t-t')}\mathbf{C}_u(t,t) \tag{3.3.0.15}$$

Notably, in the non-equilibrium steady-state regime, i.e., large t, t' and fixed lag $\tau = t - t'$, the initial condition covariance matrix equals the non-equilibrium steady-state covariance matrix, $\boldsymbol{\mathcal{V}}_0 = \boldsymbol{\mathcal{V}}_\infty$, and we are left with,

$$\mathbf{C}_u(t,t') = e^{-\boldsymbol{\Gamma}\tau}\boldsymbol{\mathcal{V}}_\infty, \qquad (\tau \geq 0). \tag{3.3.0.16}$$

Linear response function

From Eq. 3.1.1.9 and Eq. 3.3.0.2, the linear response function can be calculated as,

$$\mathbf{R}(t,t') = \left\langle \boldsymbol{u}(t)\,\boldsymbol{\eta}(t')^\top \right\rangle \mathbf{Q}^{-1} \tag{3.3.0.17}$$

$$= \int_0^t e^{-\boldsymbol{\Gamma}(t-\tau)} \left\langle \boldsymbol{\eta}(\tau)\,\boldsymbol{\eta}(t')^\top \right\rangle d\tau \mathbf{Q}^{-1} \tag{3.3.0.18}$$

$$= \int_0^t e^{-\boldsymbol{\Gamma}(t-\tau)}\, \mathbf{Q}\,\delta(\tau - t')\, d\tau \mathbf{Q}^{-1}$$

$$= \begin{cases} e^{-\boldsymbol{\Gamma}(t-t')}, & t \geq t', \\ \mathbf{0}, & t < t'. \end{cases} \tag{3.3.0.19}$$

Now we integrate the linear response function to find,

$$\chi(t,t') = \int_{t'}^t \mathbf{R}(t,s)\, ds \tag{3.3.0.20}$$

$$= \int_{t'}^t e^{-\boldsymbol{\Gamma}(t-s)}\, ds \tag{3.3.0.21}$$

$$= \boldsymbol{\Gamma}^{-1}\left(\mathbb{I} - e^{-\boldsymbol{\Gamma}(t-t')}\right), \qquad (t \geq t'), \tag{3.3.0.22}$$

where the last equality holds for all invertible $\boldsymbol{\Gamma}$.

Integral violation

Putting all our expressions together as described by the integral violation equation (3.1.2.3), we find,

$$\mathbf{I}(t,t') = \mathbf{C}_u(t,t) - \mathbf{C}_u(t,t') - \frac{1}{2}\chi(t,t')\mathbf{Q} \tag{3.3.0.23}$$

$$\begin{aligned} &= \boldsymbol{\mathcal{V}}_\infty + e^{-\boldsymbol{\Gamma}t}\,(\boldsymbol{\mathcal{V}}_0 - \boldsymbol{\mathcal{V}}_\infty)\,e^{-\boldsymbol{\Gamma}^\top t} - e^{-\boldsymbol{\Gamma}(t-t')}\boldsymbol{\mathcal{V}}_\infty - e^{-\boldsymbol{\Gamma}t}\,(\boldsymbol{\mathcal{V}}_0 - \boldsymbol{\mathcal{V}}_\infty)\,e^{-\boldsymbol{\Gamma}^\top t'} \\ &\quad - \frac{1}{2}\boldsymbol{\Gamma}^{-1}\left(\mathbb{I} - e^{-\boldsymbol{\Gamma}(t-t')}\right)\mathbf{Q} \end{aligned} \tag{3.3.0.24}$$

$$\begin{aligned} &= \left(\mathbb{I} - e^{-\boldsymbol{\Gamma}(t-t')}\right)\boldsymbol{\mathcal{V}}_\infty + e^{-\boldsymbol{\Gamma}(t-t')}\,e^{-\boldsymbol{\Gamma}t'}\,(\boldsymbol{\mathcal{V}}_0 - \boldsymbol{\mathcal{V}}_\infty)\,e^{-\boldsymbol{\Gamma}^\top t'}\left(e^{-\boldsymbol{\Gamma}^\top(t-t')} - \mathbb{I}\right) \\ &\quad - \frac{1}{2}\boldsymbol{\Gamma}^{-1}\left(\mathbb{I} - e^{-\boldsymbol{\Gamma}(t-t')}\right)\mathbf{Q}. \end{aligned} \tag{3.3.0.25}$$

In the case that $\boldsymbol{\Gamma}$ is revertible, $\boldsymbol{\Gamma}^{-1}$ commutes with $e^{-\boldsymbol{\Gamma}(t-t')}$ and we find,

$$\mathbf{I}(t,t') = \left(\mathbb{I} - e^{-\boldsymbol{\Gamma}\tau}\right)\left(\boldsymbol{\mathcal{V}}_\infty - \tfrac{1}{2}\,\boldsymbol{\Gamma}^{-1}\mathbf{Q}\right) + e^{-\boldsymbol{\Gamma}\tau}\,e^{-\boldsymbol{\Gamma}t'}\,(\boldsymbol{\mathcal{V}}_0 - \boldsymbol{\mathcal{V}}_\infty)\,e^{-\boldsymbol{\Gamma}^\top t'}\left(e^{-\boldsymbol{\Gamma}^\top\tau} - \mathbb{I}\right). \tag{3.3.0.26}$$

Moreover, if the system satisfies non-equilibrium steady-state, $\mathbf{Q}$ is uniquely given by the continuous-time Lyapunov equation (3.3.0.7) and $\boldsymbol{\mathcal{V}}_0 = \boldsymbol{\mathcal{V}}_\infty$, we get,

$$\mathbf{I}(\tau) = \tfrac{1}{2}\,\boldsymbol{\Gamma}^{-1}\left(\mathbb{I} - e^{-\boldsymbol{\Gamma}\tau}\right)\left[\boldsymbol{\Gamma}\boldsymbol{\mathcal{V}}_\infty - \boldsymbol{\mathcal{V}}_\infty\boldsymbol{\Gamma}^\top\right], \tag{3.3.0.27}$$

where it is now explicit that $\mathbf{I}(\tau) = 0$ if detailed balance holds, i.e., $\boldsymbol{\Gamma}\boldsymbol{\mathcal{V}}_\infty = \boldsymbol{\mathcal{V}}_\infty\boldsymbol{\Gamma}^\top$. Note that to compute the variance-normalized version, $\boldsymbol{\mathcal{V}}_\infty$ is transformed as specified in Eq. 3.1.2.11.

4

Results and Discussion

The primary research goal of this study is to gain understanding in how the brain's deviation from detailed balance equilibrium develops with the progression of Alzheimer's disease. As outlined in the previous two chapters, the metric we use to indicate this distance from detailed balance equilibrium is the integral violation (IV) of the fluctuation-dissipation theorem. In order to contextualize this non-equilibrium metric, we study its relation to both amyloid-beta and tau protein depositions and AD symptoms. Furthermore, we compute this metric using two different approaches: the model-free (Sec. 3.2) and the model-based analytical framework (Sec. 3.3). While not the main objective of this study, we investigate how these two approaches stack up against each other.

Firstly, in Sec. 4.1, to gain insight into how the distance from detailed balance equilibrium is changed with AD progression, we compute the average integral violation across groups per subject and per parcel (Fig. 4.1.1). Secondly, in Sec. 4.2, we investigate the relation between the IV metric and AD pathology. Concretely, we plot the average parcel integral violation with the amyloid-beta and tau depositions (Fig. 4.2.1). Furthermore, we visualize the progression of the spatial distribution of the integral violation and AD pathology (Fig. 4.2.2). In order to quantitatively investigate the spatial distribution relation, we compute the average Pearson's correlation per parcel between the integral violation and the AD pathology (Fig. 4.2.3). This is followed by a study of networks of significance along AD progression using resting-state networks (Fig. 4.2.4 and A.6.2). In the third section of this chapter, Sec. 4.3, we train a hybrid cross-validation classifier to distinguish between HC, MCI, and AD subjects (Fig. 4.3.1). Using this classifier we show that, in combination with amyloid-beta and tau values, the integral violation metric improves classification (Fig. 4.3.2 and 4.3.3**CD**). Furthermore, we highlight the top features used in classification and relate these to AD symptoms (Fig. 4.3.3**AB** and 4.3.4). Finally, in the final section, Sec. 4.4, we discuss the results, place them in the context of previous research, and mention limitations of this study.

To produce these results, we used resting-state fMRI and PET data collected by the Alzheimer's Disease Neuroimaging Initiative (ADNI). Further details on this dataset can be found in Appendix A.1. Furthermore, the exact modeling procedure, parameters, and code used to compute the results for the model-based analytical approach, are presented in Appendix A.3. Lastly, for a brief description of the statistical tests, Wilcoxon rank-sum and Monte-Carlo permutation, we refer to Appendix A.4.

4.1. Subjects and Parcels

Both FDT formalisms presented in Ch. 3, model-free and analytical, compute a scalar per parcel per subject. This gives us two options for averaging: 1) per subject; one average from 400 parcels, and 2) per parcel; one average from the number of subjects in a given group. In this analysis, the comparison groups are composed of healthy controls (HC), patients with mild cognitive impairment and heightened amyloid-beta values [MCI(AB+)], and Alzheimer patients with heightened amyloid-beta values (AD). In Fig. 4.1.1, we show the subject and parcel averages of the integral violation metric along AD progression for both the model-free and analytical approach. For both approaches, in Fig. 4.1.1**A** and 4.1.1**B**, we see a slight but statistically significant decrease in the group mean of the IV subject averages between the HC and AD groups ($p<0.05$). No statistically significant differences are found between the MCI group and the HC/AD groups. Furthermore, in Fig. 4.1.1**C** and 4.1.1**D**, parcel averages show a similar decrease in IV group mean between the HC and AD groups ($p<0.001$). Notably, there is also a significant decrease in IV group mean between the HC and MCI groups ($p<0.001$). Lastly, we see that HC parcel averages of the model-free approach are less elevated and clustered in relation to the MCI/AD averages as compared to those of the analytical approach.

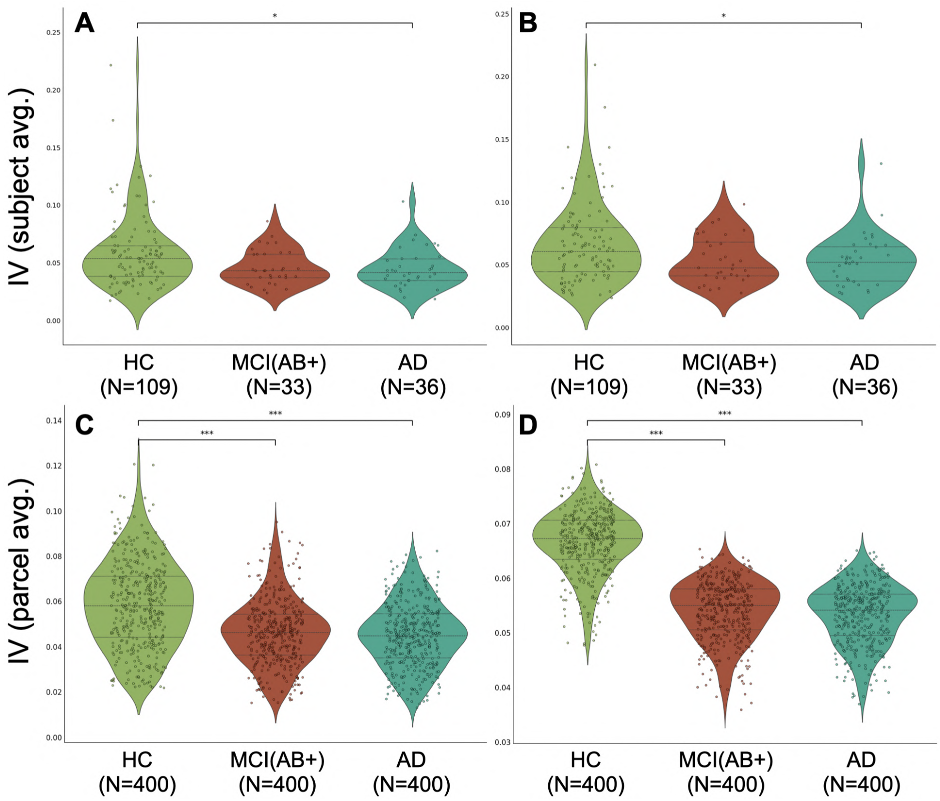

Figure 4.1.1: Integral violation and AD progression. Subject (top row; **A**, **B**) and parcel (bottom row; **C**, **D**) averages of the integral violation metric along AD progression for both the model-free (left column; **A**, **C**) and analytical (right column; **B**, **D**) approach. Number of subjects/parcels per group are given below the group label. Statistical significance between subject and parcel averages are determined by the Wilcoxon rank-sum test and the Monte-Carlo permutation test, respectively. *: p-value < 0.05, ***: p-value < 0.001.

4.2. Non-equilibrium and AD pathology

In order to place the non-equilibrium metric in the Alzheimer's disease context, we study its relation to the principal hallmark of AD pathology: the excessive aggregation of amyloid-beta plaques and tau tangles. In Fig. 4.2.1, we plot the average parcel values for IV, amyloid-beta centiloids (CL), and tau SUVR values across groups. We identify the same trend that was visible in Fig. 4.1.1**C** and 4.1.1**D**; the analytical approach (Fig. 4.2.1**B**) offers a sharper distinction between HC and MCI/AD than the model-free approach (4.2.1**A**). In general, we see that as the disease progresses, IV values decrease while and amyloid-beta CL values increase in most to all parcels. Moreover, tau SUVR scores increase along AD progression as well but in a lesser manner, a tendency identified in Fig. 1.0.1.

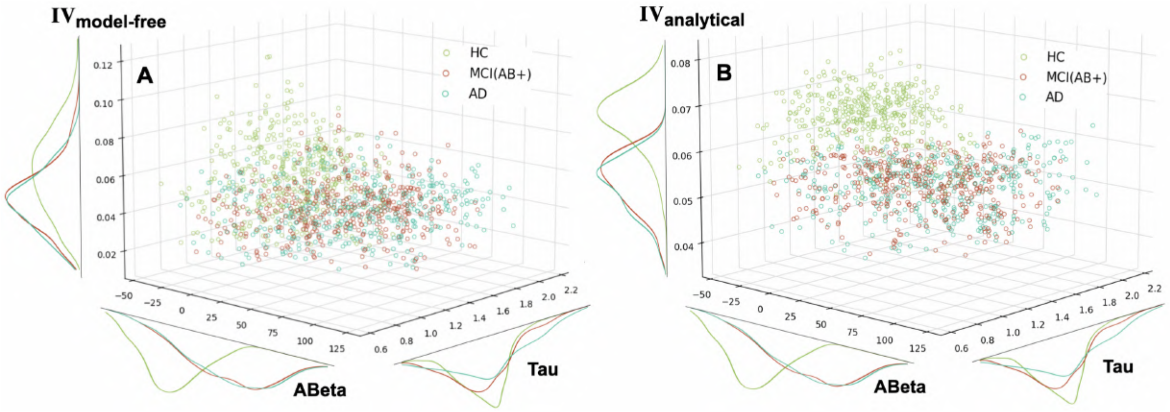

Figure 4.2.1: Integral violation and AD pathology. Parcel averages of the integral violation metric (z-axis), amyloid-beta CL (y-axis), and tau SUVR values (x-axis) along AD progression (HC, MCI, AD). Next to each axis, the normalized distributions are presented. The model-free (**A**) and analytical (**B**) IV approaches are both shown.

Crucially, to further investigate the relation between IV and amyloid-beta and tau, we look at the spatial distribution of these quantities. In Fig. 4.2.2, we visualize the distribution of the average parcel value of the IV, amyloid-beta, and tau quantities per group. Specifically, we present views of the posterior, anterior, and both laterals. In Fig. 4.2.2**A**, the model-free IV, we see that in the HC group certain regions have notably higher values of IV than others. In particular, we see elevated values near the transverse occipital, central sulcus and right temporal lobe. In contrast, the analytical IV parcel averages (Fig. 4.2.2**B**) show a much more uniform spatial distribution and decrease with disease progression. In Fig. 4.2.2**C** and 4.2.2**D**, we show the tau SUVR and amyloid-beta CL average parcel values, respectively. At first glance, we observe that while tau SUVR scores mostly increase in specific regions (i.e., temporal lobes), amyloid-beta values increase more uniformly as the disease progresses. Notably, whereas the model-free IV distribution becomes more uniform as the disease progresses, the tau SUVR scores become more region specific.

In Fig. 4.2.3, to identify a general parcel-level relation between IV and amyloid-beta and tau levels, we compute the average Pearson's correlation between these quantities per parcel for each subjects. As a benchmark, we see that the amyloid-beta and tau parcel distributions show significant correlations (mean r≈0.3) across nearly the whole brain. However, for the IV and amyloid-beta/tau correlations, significant regions become much more rare. Only

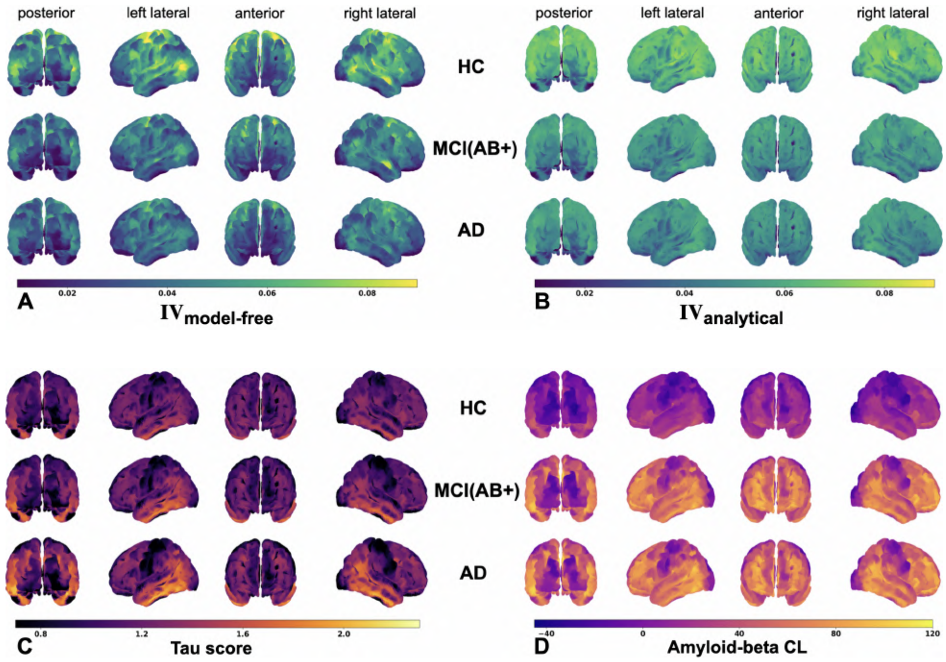

Figure 4.2.2: Integral violation and AD pathology: spatial distribution. Parcel averages of the integral violation metric (top row; **A**, **B**), tau SUVR scores (**C**), and amyloid-beta CL (**D**) along AD progression. The model-free (**A**) and analytical (**B**) IV approaches are both shown.

between 6 and 30 parcels achieve a statistically significant correlation between these quantities. Nonetheless, the overall trend of mean parcel correlation is a negative one for all combinations, which is in agreement with previous results (Fig. 4.2.1 and 4.2.2).

We know that the brain is a highly complex system that often does not allow a simple, parcel-wise deconstruction. Instead, the dynamics of a brain at rest are more reliably captured by resting-state networks (Sec. 2.2.2). In Fig. 4.2.4, we show the subject and parcel averages of the integral violation metric between HC and AD groups for each resting-state network. In Fig. 4.2.4**A**, model-free subject IV averages, we see significant decrease in the visual ($p<0.05$), salience ($p<0.01$), somatomotor ($p<0.05$), limbic ($p<0.01$), default mode ($p<0.05$) networks. Using the analytical approach, Fig. 4.2.4**B**, we lose significance for the visual, somatomotor, and limbic ($p<0.05$) networks but gain a significant decrease in the control ($p<0.05$) and default mode ($p<0.01$) network. For the parcel averages, Fig. 4.2.4**C** and 4.2.4**D**, we observe highly significant IV decreases from HC to AD across all RSNs. Again, we see that this difference is much clearer for the analytical approach than for the model-free, which is in line with previous observations in Fig. 4.1.1**CD** and 4.2.1. The MCI group is excluded in Fig. 4.2.4 as its distribution is statistically equivalent to that of the AD group (see Appendix A.6, Fig. A.6.2).

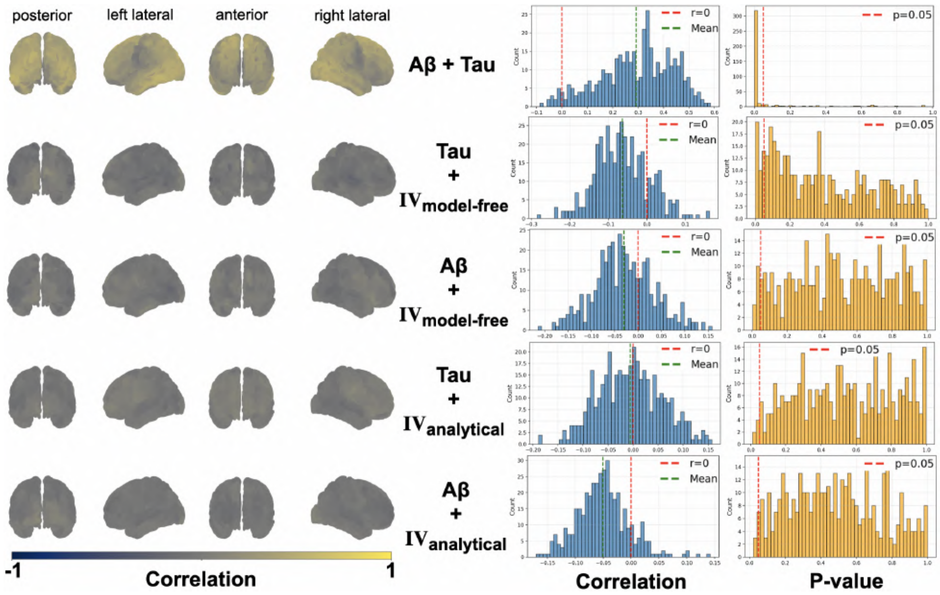

Figure 4.2.3: Integral violation and AD pathology. Average Pearson's correlations between IV, tau, and amyloid-beta per parcel for each subject. Brain plots of the posterior, anterior, and both lateral views are colored for regions with correlations between quantities (left). The blue histograms show the distribution of parcel correlations with the striped red line set on no correlation and the striped green line on the mean of the correlations. The yellow histograms show the parcel distribution of the p-values and the striped red line shows the p=0.05 cutoff. Note that in this case we can use a statistical test that uses a null hypothesis which states that the parcels are uncorrelated as the parcel values are from different, independent quantities.

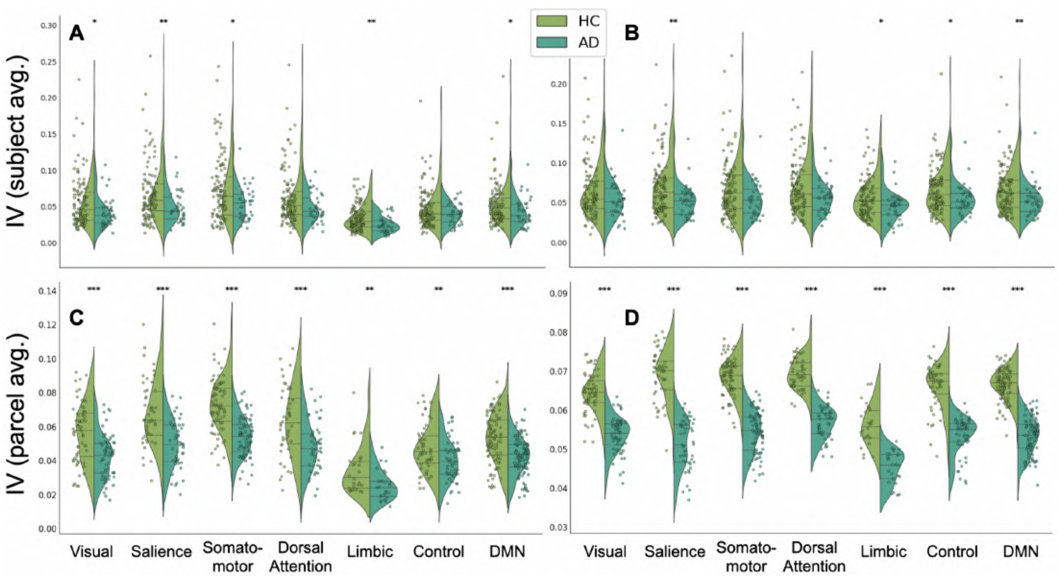

Figure 4.2.4: Integral violation and RSNs. Subject (top row; **A**, **B**) and parcel (bottom row; **C**, **D**) averages of the integral violation metric of each RSN along AD progression (HC, AD) for both the model-free (left column; **A**, **C**) and analytical (right column; **B**, **D**) approach. Statistical significance between subject and parcel averages are determined by the Wilcoxon rank-sum test and the Monte-Carlo permutation test, respectively. *: p-value < 0.05, **: p-value < 0.01, ***: p-value < 0.001.

4.3. Classification

As seen in the previous section (Fig. 4.2.2 and 4.2.3), we did not find much evidence for a direct relation between the IV metric and AD pathology. However, instead of continuing with the somewhat naive approach of trying to find more and more complex relations between quantities, we take a data-driven approach and turn towards a classification method. Generally, classification schemes allow the discovery of trends in the data that are hidden from simple, linear analysis. In this case, motivated by the seemingly low spatial correspondence between the IV metric and AD pathology, our objective is two-fold. Firstly, we set out to determine whether the inclusion of IV features adds to the classification performance. Secondly, we aim to identify areas that are especially important in the classification and relate these back to the disease.

To determine a suitable classification scheme, we look at the dimensions of the dataset. In our case, we are in a $P >> N$ setting; the dataset has a small, unbalanced sample size (HC: 109 subjects, MCI: 33 subjects, AD: 36 subjects) while containing many features per subject (3 quantities $\times$ 400 parcels). Therefore, we adopt the hybrid machine-learning framework introduced by Triebkorn et al. [53]. This approach combines random forest (RF) feature selection with support vector machine (SVM) classification, and is specifically designed for small-sample, high-dimensional settings. RFs provide interaction-aware feature pruning using an entropy criterion, while SVMs offer stable nonlinear decision boundaries once the feature space has been sufficiently reduced.

To evaluate whether IV features contribute additional predictive value, we repeat the complete procedure using three feature sets: (1) empirically derived features; amyloid-beta and tau values (800 total), (2) simulated features; integral violation values (400 total), and (3) a combined feature set (1200 total). Because the number of features far exceeds the number of subjects, we rely on strict nested cross-validation scheme as shown in Fig. 4.3.1. Following the original scheme, the number of retained features is restricted to the square root of the total dimensionality (i.e., 34). In the outer loop (100 repetitions), 25% of the subjects serve as an untouched test set. The remaining data enter the inner loop, which performs exhaustive hyperparameter optimization for both the RF and SVM. For each inner-loop split, training features are scaled using statistics solely computed from the training subset, the RF selects a reduced set of features, and the SVM is trained and validated on this selection. With 192 RF and 384 SVM hyperparameter combinations, a total of 73,728 models are evaluated per outer-loop iteration.

After the hyperparameters optimization, the inner-loop training and validation subsets are recombined, the RF performs a final feature-selection pass, and the SVM is trained using the resulting reduced feature space. In the end, classification performance is assessed on the test data.

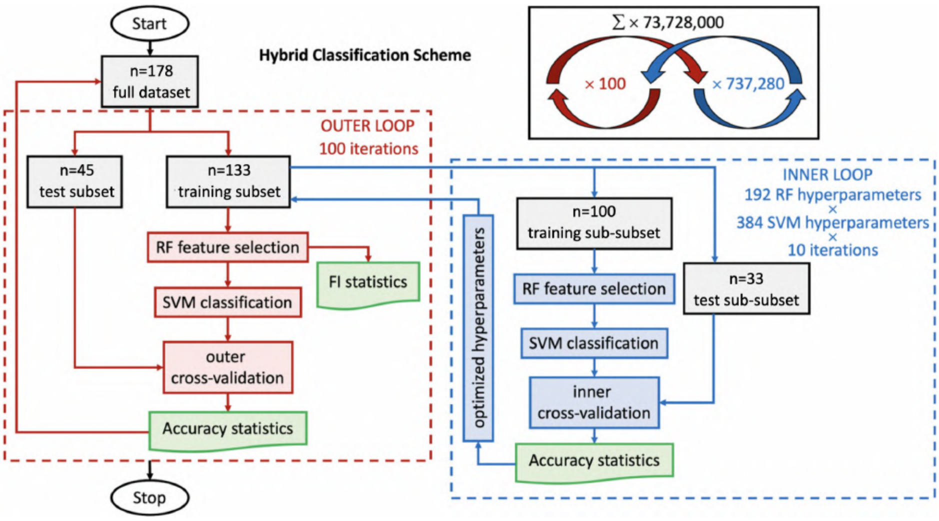

Figure 4.3.1: Nested cross-validation scheme for hybrid RF-SVM classification. In this classification scheme by [53], a random forest (RF) algorithm, for feature selection, is used together with a support vector machine (SVM), for classification. Both are used in an inner and outer loop. The outer loop performs 100 iterations with stratified 75/25 train-test splits to evaluate model generalization. Within each outer iteration, the training data enters an inner loop with 10-fold cross-validation for hyperparameter optimization. Inner loop: the data is split again. Then, both the training and test features are scaled by subtracting the median and dividing by the interquartile range. After, RF performs feature selection using entropy criterion, retaining top K features, and the SVM is trained on selected features across all hyperparameter combinations via grid search. Eventually, the optimal hyperparameters are identified based on balanced accuracy on the validation set. Back to the outer loop. Using the optimal hyperparameters from the inner loop, RF re-selects features on the full training set. Then, feature importance scores and selection frequencies are recorded. Subsequently, SVM classifies the test set using the final selected features and performance metrics (balanced accuracy, sensitivity, specificity, F1 scores) are computed. Figure adapted from [53].

Results

Overall, we used the classification scheme for three different feature sets: empirical (Aβ + Tau), simulated (IV), and combined (Aβ + Tau + IV). Additionally, we performed the classification of the simulated and combined feature sets for both the model-free and analytical approach. In Fig. 4.3.2, we present the confusion matrices for these experiments. Then, in Fig. 4.3.3, we show the top-50 feature importance distribution for the two combined feature set classifications and the mean weighted F1 scores for all five feature set classifications. Lastly, in Fig. 4.3.4, the spatial distribution of the feature importance is shown for each type of feature.

The confusion matrices of the five classification experiments (Fig. 4.3.2) indicate that the IV metric improves classification accuracy when combined with empirical features. Primarily, this is shown through the heightened classification accuracy scores for the MCI and AD subjects. For the AD subjects classification accuracy, the analytical IV approach slightly improves on the model-free approach. Furthermore, we see that the IV metric on its own leads to notably lower classification accuracies than the empirical features. Again, we observe an improvement in AD classification accuracy for the analytical compared to the model-free approach.

These observations are supported when inspecting the mean weighted F1 scores (Fig. 4.3.3**CD**).

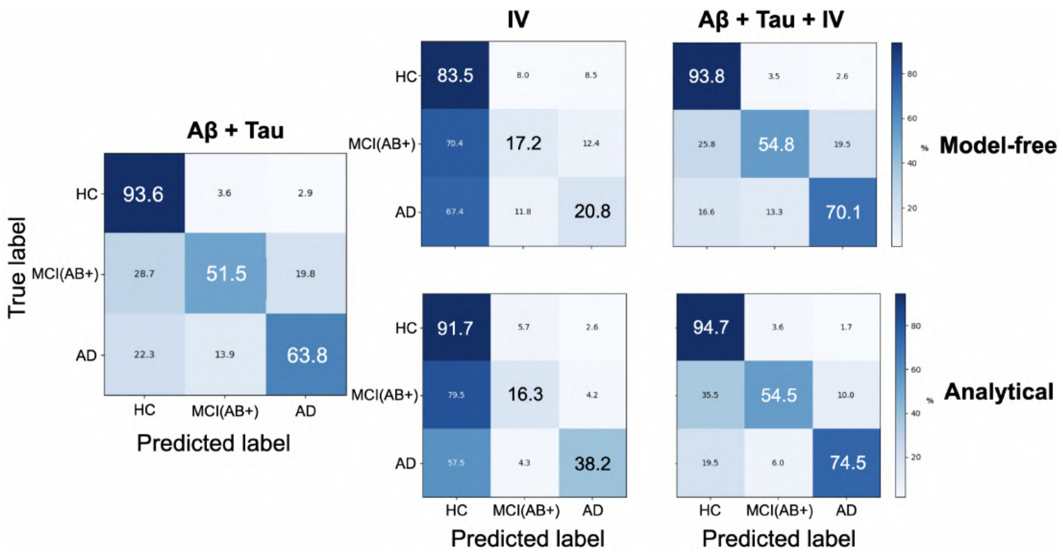

Figure 4.3.2: Normalized confusion matrices for empirical (Aβ + Tau), simulated (IV), and combined (Aβ + Tau + IV) feature set classification. The accuracy scores are computed by averaging over all 100 cross-validation runs. The model-free results are shown in the top row while the analytical results are shown in the bottom row.

Here, we see that the inclusion of the IV metric slightly but significantly increases the mean weighted F1 score by about 0.01-0.04 ($p<0.05$ for model-free, $p<0.001$ for analytical). Also, we observe that the exclusive IV feature classification achieves much lower weighted F1 scores than the empirical feature classification. Moreover, we note that the analytical IV metrics slightly increase the mean weighted F1 scores as compared to the model-free IV metrics.

In Fig. 4.3.3**AB**, we show the feature importance of the top-50 feature regions in the combined feature classification including the IV metric from the model-free and analytical approach, respectively. Additionally, in Fig. 4.3.4, the spatial distribution of these features are shown in brain renders. Firstly, we observe in Fig. 4.3.3**AB** that features with the highest importance are generally empirical features. The IV features are much less important than the empirical features. Secondly, there are more analytical IV features than model-free IV features in the top-50. In Fig. 4.3.4, we recognize some of the same regions highlighted in Fig. 4.2.2. This becomes even more evident when looking at the normalized HC-to-AD difference brain plots (Fig. A.6.1). However, most high-importance IV regions do not correspond with AD pathology regions, which is in accordance with what was shown in Sec. 4.2. Instead, high-importance IV regions are seemingly scattered more randomly across the cortex.

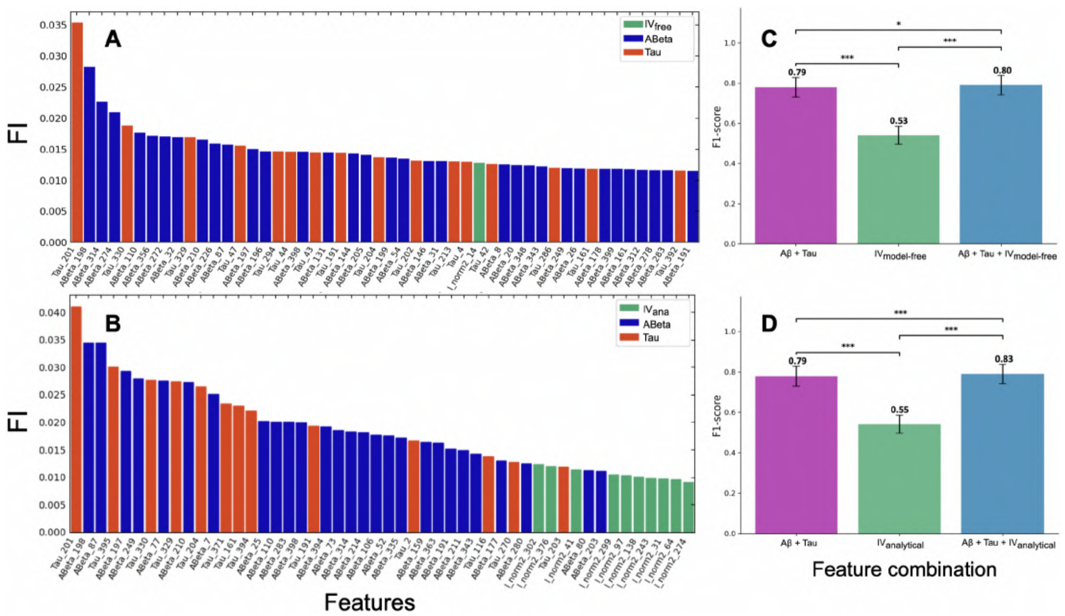

Figure 4.3.3: Feature importance distribution (left; **A**, **B**) and mean weighted F1 scores (right; **C**, **D**) for the model-free (top row; **A**, **C**) and analytical (bottom row; **B**, **D**) approach. The feature importance of IV, amyloid-beta, and tau regions are colored and numbered per Schaefer-400 parcellation [25]. The mean weighted F1 scores are shown for the three different feature combinations: Aβ+Tau, IV, Aβ+Tau+IV. Statistical significance between feature combinations is determined with the Wilcoxon rank-sum test from 100 cross-validation runs. *: p-value < 0.05, ***: p-value < 0.001.

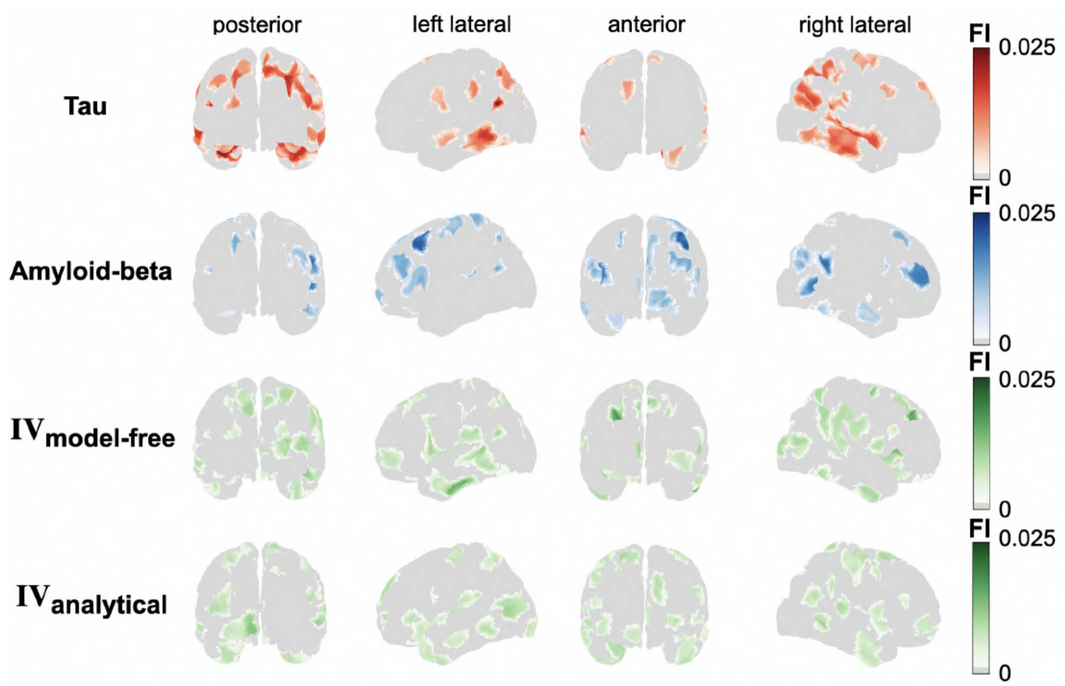

Figure 4.3.4: Feature importance (FI) spatial distribution for tau, amyloid-beta, and both IV values. The brain renders of the posterior, anterior, and both lateral views are colored for regions with top-50 feature importance for each feature type as determined in the combined classification. For the tau and amyloid-beta regions, this was done using the analytical IV combined approach.

4.4. Discussion and Limitations

In the previous sections we have seen a variety of results relating the integral violation of the detailed balance equilibrium as quantified by the fluctuation-dissipation theorem to Alzheimer's disease and its pathology. We have done this using both a model-free and model-based analytical formulation of the integral violation. In this section, we interpret these results, relate them to previous studies, and discuss various limitations.

At the whole-brain level, we found that the integral violation decreases with AD progression. To be precise, we found a significant decrease in IV between the HC and AD groups for both the model-free and analytical approach. Violations of the FDT indicate a deviation from detailed balance equilibrium. In turn, such deviations are related to information flow asymmetry between the system's components and hierarchical processes. Decreased IV in Alzheimer patients as compared to healthy controls therefore suggests that the disease reduces the information flow asymmetry and flattens the hierarchical organization in the brain. Parcel-wise distributions of the IV affirm this observation and additionally identify a significant decrease in IV between HC and MCI groups. This indicates that on the parcel level, dynamical changes become visible more quickly than on the subject level.

Furthermore, we found significantly lower levels of IV in the salience, limbic, and default mode networks of AD patients as compared to healthy controls using both the model-free and analytical approach. This suggests that these networks are disproportionately affected by the reduction in distance from equilibrium and indicates significantly flatter hierarchical organization than in the corresponding HC networks. Notably, the disruptions in the salience and default mode networks align with results from previous rs-fMRI work in AD [20]. Similar to the whole-brain level, parcel-wise averages enable more significant distinguishability than subject averages.

In general, these results agree with previous studies on information integration in AD. In particular, they affirm the trend of flatting hierarchy in AD, as made apparent by the reduced irreversibility [12] and metastability [13]. However, the increase-followed-by-decrease tendency along AD progression as found in [9] was not affirmed. Instead, on the parcel-level, we found that the MCI distribution showed an identical decrease in IV values as between the HC and AD distributions, showing no sign of increased IV due to hyperexcitability.

In order to relate the IV metric to AD pathology, we looked at the parcel-wise distributions of the IV metric and amyloid-beta and tau values. We found that as the disease progresses and amyloid-beta and tau values increase, IV values decrease. Moreover, we investigated the spatial correlation of the IV with AD pathology by computing the parcel-wise correlation for all subjects between IV, amyloid-beta, and tau values. This showed that the correlation between the spatial distribution of the IV metric and AD pathology in each subject is minimal, in particular when compared to the correlation between the spatial distributions of amyloid-beta and tau. These results indicate that although IV is a measure of AD progression, its spatial distribution is not directly related to the spreading pattern of AD pathology, suggesting that there must be unknown mechanisms or dynamics that relate the IV metric to AD pathology.

Finally, we trained a hybrid RF+SVM classifier from [53] to (1) determine whether the inclusion of IV features adds to the classification performance between HC, MCI, and AD patients, and (2) to identify regions of particular importance in the classification process, thereby potentially shedding light upon previously hidden dynamics. Overall, the inclusion of IV features significantly improved classification performance, particularly for the MCI and AD groups, evidenced by increased F1 and accuracy scores. This suggests that the IV metric, which quantifies reduced hierarchical organization, captures distinct functional information that complements traditional AD pathology features. However, it must be noted that solely IV metrics lead to significantly worse classification results than traditional AD pathology features.

To determine exactly which regions were important in classification, we looked at the mean feature importance and their visualization. Not surprisingly, for the AD pathology features, the feature importance maps reveal a correspondence with brain regions typically involved in AD progression. Tau features with the highest feature importance are predominantly located within the temporal lobes, lateral temporal cortex, and occipito-temporal gyri, regions involved in episodic memory and semantic processing and known to be susceptible to tau tangle accumulation early in the disease progression [54]. Similarly, the most important amyloid-beta features map to regions such as the frontal operculum, precuneus, and posterior cingulate cortex, which are part of the default mode and salience networks. Moreover, these networks are associated with attention, salience processing, and associative memory and are known sites of early amyloid deposition in AD [55]. These results suggest that the classification framework effectively highlights established AD pathology, with tau regions concentrated in medial and lateral temporal structures and amyloid-beta more uniformly distributed across the default mode and salience networks. This aligns with known patterns of cognitive decline in AD that link episodic memory deficits to temporal lobe tau accumulation and attentional/executive dysfunction to frontal and insular amyloid-beta deposition [20].

When looking at the high-importance IV regions, most do not correspond with AD pathology regions, which is in accordance with what was shown in Sec. 4.2. Instead, high-importance IV regions are seemingly scattered across the cortex. However, we find that a substantial number of these regions are associated with clinical manifestations of AD. Indeed, most high-importance IV regions map onto hubs of the default mode network, including the posterior cingulate cortex, precuneus, medial prefrontal cortex, and lateral parietal areas. Disruption of default mode network's connectivity, in particular in the posterior cingulate and precuneus, has been linked to early AD and correlates with impairments in episodic memory and self-referential processing [11]. At the same time, the heightened feature importance of regions in the medial and lateral temporal cortical regions is consistent with known symptom progression in the medial temporal lobe, which is a major determinant of episodic memory dysfunction and cognitive decline in AD [56]. Furthermore, several IV features also fall within prefrontal and orbitofrontal cortices, regions known to exhibit metabolic dysfunction in AD, thereby contributing to executive impairments and behavioral changes as the disease progresses [57]. Moreover, the appearance of regions belonging to the salience and frontoparietal control networks aligns with the broader pattern of network disintegration characteristic of AD, which underlies deficits in attention and cognitive control [2]. However, it must be noted that various high-importance IV regions are not associated with known AD hot spots, such as the visual and somatomotor networks.

Nonetheless, taken together, these results indicate that the IV metric captures several functional alterations in AD subjects associated with memory and other related cognitive symptoms. Which suggests that locally flattened hierarchy in particular brain regions correlates to reduced cognitive performance in these regions. Crucially, these outcomes affirm a wider trend found in FDT studies, namely that the strength of hierarchical processes is positively related to cognitive complexity, as shown in [15], [16], [23].

Throughout the thesis, the IV metric was computed using both the model-free and model-based analytical implementation. The close agreement between the two confirms the reliability of the approaches and speaks to the validity of the IV measure, as was also found in the original implementation [9]. However, in some results differences did appear. In particular, we saw that the HC parcel averages of the model-free approach are visually less different and clustered in comparison to those of the analytical approach. Notable, however, is that even the model-free approach was able to confidently distinguish between these groups; no significant difference was gained. The same cannot be said for the RSN analysis. Although the general trend is similar, both methods identify significant differences in salience, limbic, and default mode networks, the results do change significantly. Generally, suggesting that the analytical approach is able to use its advantage of being able to obtain an exact, model-based estimation of steady-state dynamics to improve distinguishability at the parcel level.

Further proof can be found in the spatial distribution of the IV metric. Here, we observed a more uniform spread of values using the analytical approach as compared to the model-free approach. Which, again, potentially indicates the increased robustness of the model-determined steady-state dynamics.

Using the classifier, we found that the analytically computed IV metric slightly outperformed the IV metric that was determined by the model-free approach in both F1 and accuracy scores. Predominantly, the analytical approach improved the AD group classification. Moreover, more analytical than model-free IV features show up in the top-50 important features using the combined classification method. Further indicating that analytical IV features provide more additional information than model-free IV features. Finally, many high-importance IV features that are linked to cognitive impairments related to AD overlap for both the model-free and analytical approaches. Suggesting that although the analytical IV is able to access more robust steady-state dynamics, the model-free approach is similarly capable of identifying regions of flattened hierarchy and subsequent reduced cognitive ability.

Limitations

Behind and beyond the results interpreted above, there are various aspects that limit their applicability and validity. Some of these are framework specific, while others are inherent to the brain models and imaging data. In this part, we highlight and discuss selected aspects.

A primary aim of this study is to improve our understanding of how non-equilibrium dynamics are manifested in AD. A crucial part in validating the FDT framework's addition to this understanding is demonstrating its usefulness in diagnosis and predicting disease course progression. In particular, we are interested in distinguishing patients at stages that are difficult to diagnose with current methods, such as pathological markers or clinical evaluations, that is, the MCI stage. Although we saw significant differences between HC and AD groups on subject and parcel level, we observe no significant difference between MCI and

AD groups. Furthermore, even though the classification performance of MCI patients was improved with the addition of our non-equilibrium metric, this improvement was minimal. Ultimately, the IV metric has not shown the capability to extract truly substantial information for MCI distinguishability from the resting-state fMRI data.

Clues to why the FDT framework is unable to make a substantial difference in this area might be sought in the spatial distribution of the IV metric. We found that distributions correlate minimally with amyloid-beta and tau deposition distributions. Furthermore, although the classifier was able to highlight IV features that can be related to AD symptoms, indicating that the IV metric is functionally associated with the disease, it has not been able to do so in a reliable and statistically significant manner. Taken together with the observation that at the whole-brain level, correlations were found between IV and AD pathology, these results suggest the existence of an underlying, and yet unknown, mechanism connecting these phenomena. Thus, despite hints of possible involvement, it has proven difficult to robustly relate the FDT framework to AD pathology.

Moreover, even though the overall decline of non-equilibrium dynamics in AD aligns with the trend relating hierarchical organization with active and healthy cognition as identified in previous work, there is a point of contention: hyperexcitability in the MCI stage. Whereas Patow et al. [9] showed an increase in non-equilibrium dynamics as compared to HC, our results indicate the opposite, at least on parcel level. This suggests that the FDT framework is not perfectly consistent in identifying the non-equilibrium alternations of the MCI stage, which fits the earlier observation of limited applicability to MCI patients. As a consequence, we are currently unable to comment on the effect of hyperexcitability in a definitive manner using the FDT framework.

Possible limitations on the technical side include inherent constraints and restrictions associated with brain modelling and neuroimaging data. Firstly, we note that the linear nature of the model used in this study, the linear Hopf model, is both its strength and main limitation [8]. Although the computation of the model's statistics is much faster, only linear dynamics are captured, which have been shown not to portray the complete dynamical picture [58]. Furthermore, whereas the Hopf model and most other node-based models only incorporate pairwise interactions, higher-order interactions have recently been shown to carry importance for analyzing integrative processes in the brain using fMRI data [37].

Moreover, to reliably extract the steady-state dynamics from fMRI data, the time series require a minimal length [59]. In the ADNI dataset used in this study, the minimal length is 197 time points. Arguably, for the purpose of extracting robust steady-state dynamics, this is on the shorter side, possibly impacting result reliability. Additionally, the repetition time of the fMRI scans, the time between each full brain scan, is 3 seconds. This is on the longer side of the fMRI spectrum and impacts the model's ability to capture the temporal aspects of the dynamics. Plausibly, this becomes apparent through the model fitting statistics. Although the MSE values and FC correlations between empirical data and simulation are reasonable for the used parcellation, COVτ correlations are mediocre and further indicate that the model is only moderately able to capture the temporal dynamics of the fMRI data, deteriorating the model's ability to capture robust steady-state dynamics and impacting result accuracy.

Lastly, to fit the effective connectivity with biological plausibility, the structural connectivity is used as guiding factor [41]. However, in our dataset not every participant's SC was determined. Therefore, we used a combination of the empirical FC and COVτ matrices as initial condition. This undoubtedly affects the biological plausibility of our EC matrices and thus, our model-based approach.

5 Conclusion and Outlook

In this thesis, we investigated Alzheimer's disease from a dynamical and functional perspective by quantifying deviations from detailed balance equilibrium using the fluctuation–dissipation theorem. Using both a model-free and linear Hopf model-based analytical implementation of the integral violation metric, we consistently found that Alzheimer's disease is associated with a reduction in distance from equilibrium.

At the whole-brain level, integral violation decreases significantly from healthy controls to Alzheimer patients, indicating reduced information flow asymmetry and a flattening of hierarchical organization. Parcel-wise and network-level analyses showed that these dynamical alterations are more pronounced at node level and disproportionately affect the salience, limbic, and default mode networks, which are associated with memory and other Alzheimer-related cognitive symptoms. These findings align with previous observations of reduced irreversibility and metastability in Alzheimer's disease and support the characterization of the disease as a disorder of impaired information integration.

Although integral violation decreases with increasing amyloid-beta and tau burden at the group level, its spatial distribution shows minimal correspondence with protein deposition patterns. This mismatch suggests that non-equilibrium dynamics capture functional alterations that are not directly explained by local pathology. Consistent with this interpretation, integral violation features improve classification performance when combined with amyloid-beta and tau values, particularly for MCI and Alzheimer patients, indicating their complementary value. Furthermore, many high-importance integral violation regions, identified by a hybrid RF + SVM classifier, are associated with clinical manifestations in Alzheimer patients, confirming the framework's ability to connect brain dynamics with cognitive impairment.

The close agreement between the model-free and analytical implementations confirms the similarity of the methodologies and speaks to the validity of the integral violation metric. At the same time, the analytical approach provided improved parcel-level distinguishability and MCI classification performance, likely due to its more exact characterization of steady-state dynamics. Overall, these results support the view that Alzheimer's disease is characterized by a progressive loss of non-equilibrium through flatter hierarchical organization and establish the distance from equilibrium metric as a meaningful indicator of cognitive function.

Outlook

The research presented in this thesis can be continued and improved upon in various aspects. In this section, we briefly identify several important directions that remain available for future investigations.

To start, the minimal spatial correspondence between IV and amyloid-beta or tau deposition suggests the involvement of unknown, underlying mechanisms that link pathology to non-equilibrium dynamics. Future studies could explicitly model the protein and neurodegeneration spreading patterns and study how these processes relate to non-equilibrium deviations using longitudinal data. This would provide mechanistic insights into how pathology and non-equilibrium dynamics are related in the AD brain while contributing to a larger trend of relating structure to function using PET and fMRI data.

These more intricate studies have only limited feasibility with the current ADNI-3 dataset. Instead, we would commend applying the presented FDT framework on longer (i.e., more time points) and faster (i.e., shorter repetition time) fMRI datasets, preferably containing SC data for increased biological plausibility, while using a cruder data parcellation to improve the estimation of robust steady-state dynamics through better model fitting and test robustness across datasets. Such an approach would simultaneously increase the model's ability for simulating temporal dynamics and test the generalizability across datasets. As accurate results of the FDT framework are hypothesized to rely on robust estimations of steady-state dynamics, a future study could investigate this by combining such an fMRI dataset with a more detailed analysis of the approximation of steady-state dynamics.

Again, looking at the bigger picture, the IV metric provides only one approach to characterizing non-equilibrium dynamics. A future, meta-study could combining the present framework with other non-equilibrium and hierarchical measures such as the intrinsic ignition framework [60] or trophic levels approach [61], allowing for the inclusion of multiple angles on hierarchical organization and its degradation in Alzheimer brains. Connected to this would be the application of different brain models. As the analytical FDT framework is compatible to all models that adhere to Langevin dynamics and are able to simulate an EC matrix, the possibilities are quite unrestricted. Perhaps the most interesting would be to apply and compare models that include non-linear dynamics and/or higher-order relations between nodes to determine the amount of non-equilibrium hidden in these interactions [36].

Furthermore, without restricting ourselves to one imaging modality, we suggest extending the framework to other neurophysiological modalities. Applying FDT-based non-equilibrium measures to EEG or MEG data would allow the study of faster timescale dynamics and could provide additional insights into excitation–inhibition balance and hyperexcitability in early disease stages. Multimodal comparisons may further clarify how non-equilibrium properties manifest across different spatial and temporal scales. Moreover, we note that various M/EEG databases are readily available for AD analysis. Notably, multimodal implementation would involve taking another critical look at data normalization, an important topic in the future development of the analytical FDT framework (*J.Monti, personal communication, Dec. 2025*).

Lastly, beyond relating structure to dynamics, we suggest investigating the relationship between the IV metric and clinical disease severity. While this thesis' analyses were primarily aimed at relating the IV metric with AD pathology, a natural next step would be to relate the IV metric to cognitive scores, such as MMSE or MOCA. Thereby examining whether reduced distance from equilibrium directly corresponds to clinical impairment, providing a more robust link between dynamics and cognitive performance. Moreover, such analyses could clarify the observed difficulties with MCI subjects and test the IV metric as functional biomarker or symptom predictor. Additionally, such data is already included in the ADNI database, which guarantees feasibility.

A

Appendix

A.1. Data and Preprocessing

The results in this thesis are determined using the third phase dataset [10] from the Alzheimer's Disease Neuroimaging Initiative (ADNI), a large, longitudinal study designed to track the progression of Alzheimer's disease. This ADNI-3 dataset contains neuroimaging data, such as PET scans highlighting amyloid-beta and tau proteins and resting-state fMRI (repetition time: 3 seconds, minimal length: 197 time points), and cognitive examination data of 238 participants. The main demographic and clinical characteristics are summarized in Tab. A.1.1. The analysis of this thesis only uses HC, and MCI + AD participants with heightened amyloid-beta values, i.e., MCI(AB+) and AD(AB+). Those participants that are clinically classified as MCI or AD but without heightened amyloid-beta values were excluded from this study. The reason being that it could be reasonably argued that these participants suffer from a non-AD form of dementia.

After the data was acquired by ADNI, Noelia Martinez and David Aquilué performed the preprocessing steps and we decided on a suitable parcellation: the Schaefer-400 [25]. The main considerations when choosing a parcellation are data availability, grouping criteria, and computational cost. Given that there are more data available with only cortex, we restricted ourselves to a cortical parcellation. Then, as resting-state networks are often used in the literature, we intended to use them as well. Therefore, we were left with the Schaefer parcellations [25], as these are based on the Yeo7 networks [34]. Finally, we decided on using 400 regions as it gives a level of detail that allows us to relate our metric to the AD pathology of specific regions. Additionally, our computational model, the linear Hopf model [8], is efficient enough to not be hindered too much by the size of the parcellation.

Preprocessing

Resting-State fMRI Preprocessing
The rs-fMRI data preprocessing adhered to the standard volume-based pipeline implemented within the Matlab-based CONN toolbox (v. 22v2407, `http://www.nitrc.org/projects/conn`), consistent with previously published methodology [43]. The standard pipeline comprised the following sequential steps: functional realignment for motion correction;

Table A.1.1: Demographic and clinical characteristics for the ADNI-3 dataset (N=238 participants, outliers removed). Means ± SD are displayed for continuous measures. Absolute values are displayed for categorical measures. Abbreviations: MMSE Time = Mini-Mental State Examination – Orientation to Time; MMSE Place = Mini-Mental State Examination – Orientation to Place; MMSE Memory = Mini-Mental State Examination – Memory (Registration and Recall); MMSE Overall = Mini-Mental State Examination – Total Score; MOCA = Montreal Cognitive Assessment; FAQ = Functional Activities Questionnaire; NPI = Neuropsychiatric Inventory; CDGlobal = Clinical Dementia Rating – Global Score; GDTotal = Geriatric Depression Scale – Total Score.

	HC (n=109)	**MCI (n=90)**	**AD(n=39)**	**Test**	**p-value**	**Effect size**
Age	73.39±5.83	73.96±6.38	75.44±7.21	Kruskal-Wallis	0.059	
Genetic Sex (M/F)	52/57	48/42	27/12	Chi-square	0.069	
MMSE Time	4.88±0.33	4.62±0.61	3.21±1.36	Kruskal-Wallis	< 0.001	0.327 (ε^2)
MMSE_Place	4.74±0.48	4.46±0.62	3.79±1.17	Kruskal-Wallis	< 0.001	0.137 (ε^2)
MMSE Memory	9.62±0.61	9.08±0.91	7.00±1.89	Kruskal-Wallis	< 0.001	0.336 (ε^2)
MMSE Overall	28.93±1.18	27.48±1.93	22.46±2.96	Kruskal-Wallis	< 0.001	0.490 (ε^2)
MOCA	25.58±2.45	22.24±3.36	15.41±4.14	Kruskal-Wallis	< 0.001	0.500 (ε^2)
FAQ	0.38±1.48	2.97±3.86	15.90±6.89	Kruskal-Wallis	< 0.001	0.584 (ε^2)
NPI	0.58±2.48	3.70±6.24	11.03±9.90	Kruskal-Wallis	< 0.001	0.354 (ε^2)
CDGlobal	0.01±0.07	0.49±0.12	0.85±0.37	Kruskal-Wallis	< 0.001	0.887 (ε^2)
GDTotal	0.80±1.22	1.85±1.78	2.10±1.62	Kruskal-Wallis	< 0.001	0.149 (ε^2)

slice-timing correction to account for temporal acquisition differences; identification of outlier scans via the Artifact Detection Toolbox (ART) (`http://www.nitrc.org/projects/artifact_detect`); coregistration with the participant's high-resolution T1-weighted image; spatial normalization to the MNI-152 standard space with a 2mm isotropic resampling resolution; and final smoothing with an 8mm FWHM Gaussian kernel.

Noise reduction was performed using the anatomical CompCor (aCompCor) method [62], as implemented in the CONN toolbox. This procedure involved regressing out several sources of noise from the functional data: the first five principal components attributable to each individual's white matter signal; the first five components from the cerebrospinal fluid signal; the six subject-specific rigid-body realignment parameters and their first-order temporal derivatives; the artifacts identified by ART; and the main effect of the scanning condition. Following denoising and linear detrending, the BOLD signal timeseries were band-pass filtered within the 0.008–0.08 Hz range to remove both low-frequency drift and high-frequency noise.

PET Preprocessing

Prior to analysis, the raw PET data were co-registered to the first scan and averaged, with poor-quality scans discarded by the source's PET Core Group. Preprocessing was performed using SPM12 and included the co-registration of each PET image to the participant's native T1-weighted MRI. The deformation fields obtained from the T1 segmentation step (detailed in the Structural MRI section) were then used to normalize the PET scans to the MNI-152 standard space at 2mm isotropic resolution. Based on the scan metadata, the normalized PET images were smoothed with an 8mm FWHM Gaussian kernel.

To minimize spill-in and spill-out effects, Partial Volume Correction (PVC) was applied after the co-registration step using the Müller-Gärtner algorithm, implemented in the PETPVC toolbox [63]. PET scans for Tau protein tracers (Flortaucipir) were converted to SUVR using the inferior gray cerebellar as reference.

All PET scans for amyloid tracers, Florbetapir (FBP) and Florbetaben (FBB), were converted to Standardized Uptake Value Ratios (SUVRs) using the whole cerebellum as the reference region [64], [65]. Subsequently, both FBP and FBB SUVRs were converted to the harmonized Centiloid (CL) units using specific calibration equations (FBB: $CL = 94.082 \times SUVR$˘94.772; FBP: $CL = 110.982 \times SUVR$˘116.172). This conversion followed the GAAIN Centiloid Project Level-1 and Level-2 procedures [66]. Finally, amyloid status was determined using a global cutoff of 24 Centiloids (equivalent to SUVR 1.11 for FBP and 1.08 for FBB) [67].

A.2. The BOLD effect and Neural activity

In order to relate the BOLD effect to neural activity, we describe two distinct relations. That is, the link between the BOLD signal and metabolic activity and the relation between metabolic activity and neural activity.

We start from the most well-known limitation of BOLD-based imaging; the neural-metabolic delay. Early studies of BOLD-fMRI noted that when a stimulus is applied, evoking a change in neural activity, the change in BOLD signal lags behind the stimulus onset and continues after stimulus termination [68], [69]. The delays differ in duration, the initial BOLD response generally occurs in the 2-6 seconds range, whereas the return to baseline is completed in roughly 20 seconds after stimulus termination [28]. A model of these observations is given through the hemodynamic response function (HRF), a quantification of the BOLD response after a neural stimulus. In mathematical terms, the signal response $I(t)$ is the convolution of the stimulus pattern $X(t)$ and the HRF $h(t)$,

$$I(t) = X(t) \circledast h(t). \tag{A.2.0.1}$$

The one shown in Fig. A.2.1 is known as the canonical HRF and is widely used in analysis programs such as SPM [28].

Most HRFs are composed of three subsequent parts: the initial dip, the ramp, and the undershoot. The most pronounced of these is the ramp phase. As mentioned before, the increase in MR signal arises, after neural activation, from a drop in concentration of deoxyhemoglobin due to a more substantial increase in CBF than in CMRO_2. A common explanation of this delay is through the non-instantaneous response of the CBF. The magnitude of response is closely related to the CBF increase from baseline, the ratio n of fractional change in CBF compared to the fractional change in CMRO_2, and the scaling factor M, which is related to the amount of deoxyhemoglobin present in the baseline. Before the onset of the ramp and subsequent main response, there is an initial dip in signal. Unlike heart and skeletal muscle, brain tissue lacks local oxygen reserve. Therefore, just after a sudden increase in neural activity, an immediate rise in CMRO_2 is expected and occasionally observed [27]. The first studies of this initial dip were somewhat conflicting in their observations [70], though improved temporal precision and sensitivity in MR acquisition methods have made the phenomenon more established [71]. Nonetheless, many previous generation MRI scanners encounter significant trouble in identifying the initial dip due its relatively weak response and this is reflected in many of the available datasets and corresponding HRFs.

After the initial dip and subsequent ramp phase, the MR signal decreases slightly beyond baseline; the undershoot. The main hypothesis is that the blood volume settles back to baseline more slowly than the CBF. This is described through the balloon model, where the venous compartment is modeled as an expandable balloon [26]. At steady state, the inflow rate F_{in} and outflow rate F_{out} are equal, $F_{in} = F_{out}$. However, when neural activity leads to a CBF increase, $F_{in} > F_{out}$ and the balloon inflates. As the CBF is the driving function of the system, we consider the inflow rate as a function of time $F_{in}(t)$ and the outflow rate as a function of the balloon volume $F_{out}(v)$. Now, the outflow rate is dependent on the internal pressure of the balloon, which is proportional to the balloon inflation. This makes the curve of $F_{out}(v)$ analogous to the stress-strain curve and dependent on the biophysical properties of

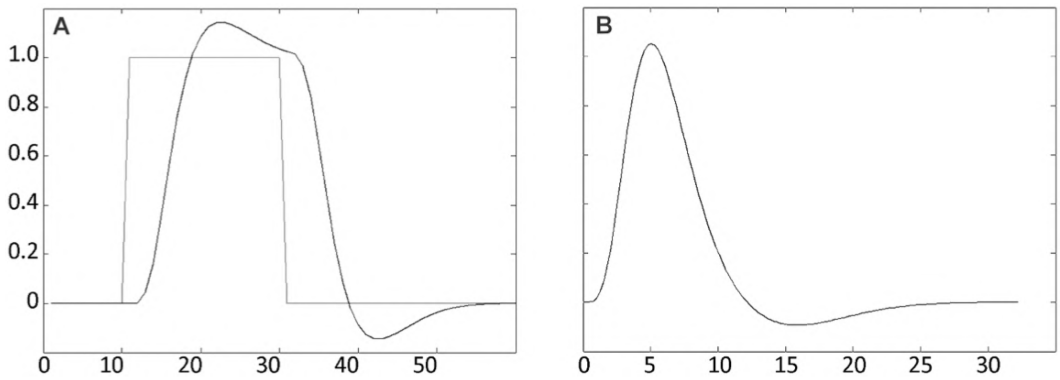

Figure A.2.1: The BOLD response $I(t)$ in arbitrary units over time in seconds to a (A) block stimulus $X(t)$ and (B) "delta" stimulus $X(\delta(t = 0))$; the canonical hemodynamic response function $h(t)$. Observe the ramp phase shortly after stimulus and the post-stimulus undershoot. Also, note the lack of initial dip. Adapted from [28].

the balloon. This model has been shown to closely match observed response curves without the need for unsupported CBF undershoots.

Now that we have related the metabolic signatures (e.g., CBF, CMRO_2, and volume) to the BOLD signal through the HRF, we need to describe the relation between these signatures and the underlying neural activity; the neurovascular coupling. As these connections are part of an active research area, there is only a limited understanding of neuron and glial cell energy metabolism. In the last decade it has become clear that astrocytes play a key role in controlling the local cerebrovascular microcirculation. When neurons are activated, pre-synaptic input is increased and neurotransmitters are released into synaptic clefts. Subsequently, astrocytes perform various important functions that increase energy demand. This demand is met through both oxidative and non-oxidative metabolic processes. The exact distribution and characteristics of these processes is not definitively known yet. However, there are various theories on how both mechanisms underlie neurovascular coupling [28]. From studies combining BOLD-fMRI and conductivity measures we have seen that hemodynamic changes correlate more with local field potentials (LFPs) than with single-unit or multi-unit recordings. In addition, a strong coupling between LFPs and tissue oxygen concentrations has been shown in absence of neural spiking, further supporting the notion that (pre-)synaptic mechanisms, rather than spiking activities, more strongly drive the metabolic machinery [72].

Related to the neurovascular mechanisms is the impact of disease and other sources of variation on the baseline values of the metabolic signatures. As noted before, the strength of the BOLD signal is at least in two ways related to baseline values: (1) the CBF and (2) the M factor. This last factor represents the maximum possible BOLD signal change that can be achieved in a given voxel or brain region and sums up a combination of underlying mechanisms. An empirical method of determining these baseline values is by constructing a calibrated-BOLD model through the use of arterial spin labeling (ASL) [27]. Captured in this model are the physiological impairments to the neurovascular response.

Here, we briefly discuss age-related and AD-related impairments. Ideally, we would describe how AD pathology influences neurovascular components and relate this to the observed signals. However, in the absence of both a complete AD pathology and unified neurovascular theory, we divert our efforts to reviewing empirical evidence. In general, ageing is related with many alterations in the brain, such as neurodegeneration, synaptic decline, and metabolic changes. In particular, the medial temporal lobe (MTL) shows a particular vulnerability to these processes. The MTL includes areas such as the hippocampus and amygdala and functionally shows increased activity during memory processing [73]. In comparison to younger adults, several studies have reported increased BOLD signals in the MTL during memory retrieval tasks in older adults [22]. However, as other studies show an overall lowering of signal magnitude with an increase in age [74], caution is advised when comparing neurovascular measures across age groups. Regarding AD-related alternations, compared to normal ageing, patients with AD showed a greater and steeper reduction in oxygenated hemoglobin concentrations as the disease severity progressed [75]. Meanwhile, as mentioned in Sec. 1, other studies point to a heightened BOLD response (i.e., hyperexcitability) in AD patients compared to HCs [22].

A.3. Model parameters, Code availability, and AI use statement

In this part of the appendix, we describe the exact modeling procedure and present the fit statistics in more detail. Additionally, we provide a link to the GitHub repository that includes the code used to produce the results presented in this thesis. Lastly, we give a short description of how AI was used in this study.

Model parameters and fit statistics

An overview of the model-based approach is presented in Fig. A.3.1. The brain model used in this study is the linear approximation of the Hopf/Stuart-Landau model as described in Sec. 2.3. We briefly walk through the modeling procedure.

All of the steps are done on subject basis. First, we extract the intrinsic node frequencies from the fMRI data by computing the discrete fourier transfrom (Fig. A.3.3**EF**). Second, we apply a global z-scoring and detrending of the time serie data and afterwards calculate the empirical FC and COVτ. Third, we initialize the EC by a linear combination of these empirical matrices. Also, we initialize the node values of the bifurcation parameter and noise parameter at -0.02 and 0.45, respectively. Fourth, we use the gradient-based Adam optimizer to iteratively fit the EC (Fig. 2.2.2, bifurcation parameter per node (Fig. A.3.3**AB**), and noise parameter per node (Fig. A.3.3**CD**). In each optimization loop we compute the simulated FC and COVτ using the Hopf model, evaluate the subsequent loss function, Eq. 2.3.2.7, and compute the gradients for each fitting module. Notably, we explicitly did not allow negative values in the EC as this would break the analytical FDT. Fifth, when the optimizer cannot find a better solution, we compute and save the final EC, bifurcation parameters, and noise parameters. Also, we run the loss function one last time to compute the final loss metrics. The hyperparameters used in this study are presented in Tab. A.3.2.

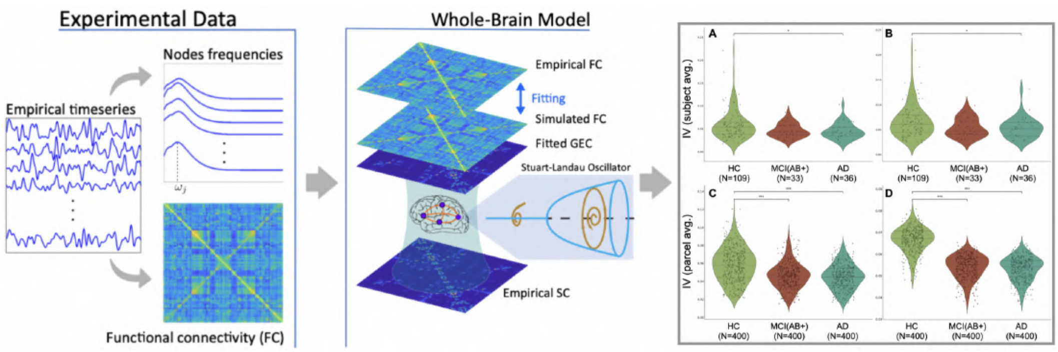

Figure A.3.1: Overview of the whole-brain modeling pipeline. The empirical measures such as the top node frequencies and BOLD-fMRI time series are used to fit a model of Stuart-Landau oscillators using the functional connectivity and delayed covariance matrix. From the simulated effective connectivity and fitted parameters, out of equilibrium metrics are computed using the analytical FDT framework (Sec. 3.3). Adapted from [16].

The final all-subject average fitting statistics are presented in Tab. A.3.1. We observe that the FC correlations and MSE values are reasonable. However, the low COVτ correlation implies that the model is not able to accurately capture the temporal relations. Consequently, this could have mean that the steady-state dynamics are not very well characterized, thereby affecting the final FDT results.

Table A.3.1: Average fitting statistics: Pearson correlation and mean squared error between empirical and simulated FC and COVτ matrices.

Fit statistic	Mean and standard deviation
$\mathrm{Corr}_{\mathrm{FC}}$	0.50 ± 0.04
$\mathrm{Corr}_{\mathrm{COV}}\tau$	0.20 ± 0.09
$\mathrm{MSE}_{\mathrm{FC}}$	0.050 ± 0.009
$\mathrm{Corr}_{\mathrm{COV}\tau}$	0.034 ± 0.006

Note that for the model-free approach, we applied the same z-scoring and detrending as done here. Also, to determine an appropriate τ value for the COVτ matrices, we compute the average auto-correlation of both empirical and simulated covariance matrices. We found that $\tau = TR$ gave the closest correspondance to the $1/e \approx 0.37$ value.

Furthermore, we remark that previous Hopf model implementations did not fit the bifurcation and noise parameters [15], [16]. We decided for that approach to improve fitting statistics. Due to the scope and time constraints of the project, no other methods were used and subsequently compared with. This implies that additional investigation is warranted in this section.

Table A.3.2: Linear Hopf model hyperparameters

Parameter	Value
Loss weight (MSE FC)	0.10
Loss weight (MSE covariance at τ)	0.10
Loss weight (Correlation FC)	0.40
Loss weight (Correlation covariance at τ)	0.40
Learning rate (EC)	1×10^{-4}
Learning rate (σ)	1×10^{-4}
Learning rate (a)	5×10^{-4}
Adam β_1	0.9
Adam β_2	0.999
Adam ϵ	1×10^{-8}
Maximum iterations	1000
Iteration check interval	50
Early stopping patience	5
Maximum coupling (C_{max})	0.2
Initial value (σ)	0.45
Initial value (a)	-0.02
Repetition time (TR)	3 s
Global coupling (g)	1
Time constant (τ in TR)	1.0

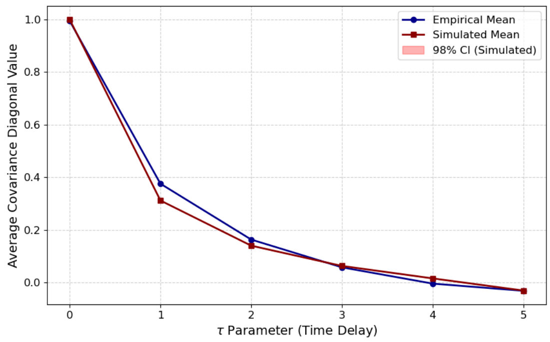

Figure A.3.2: Average auto-correlation values for $N = 60$ versus τ delay (in TR) for both the empirical and simulated covariance matrices. Note that the auto-correlation drops below $1/e \approx 0.37$ after $\tau = TR$ for both matrices.

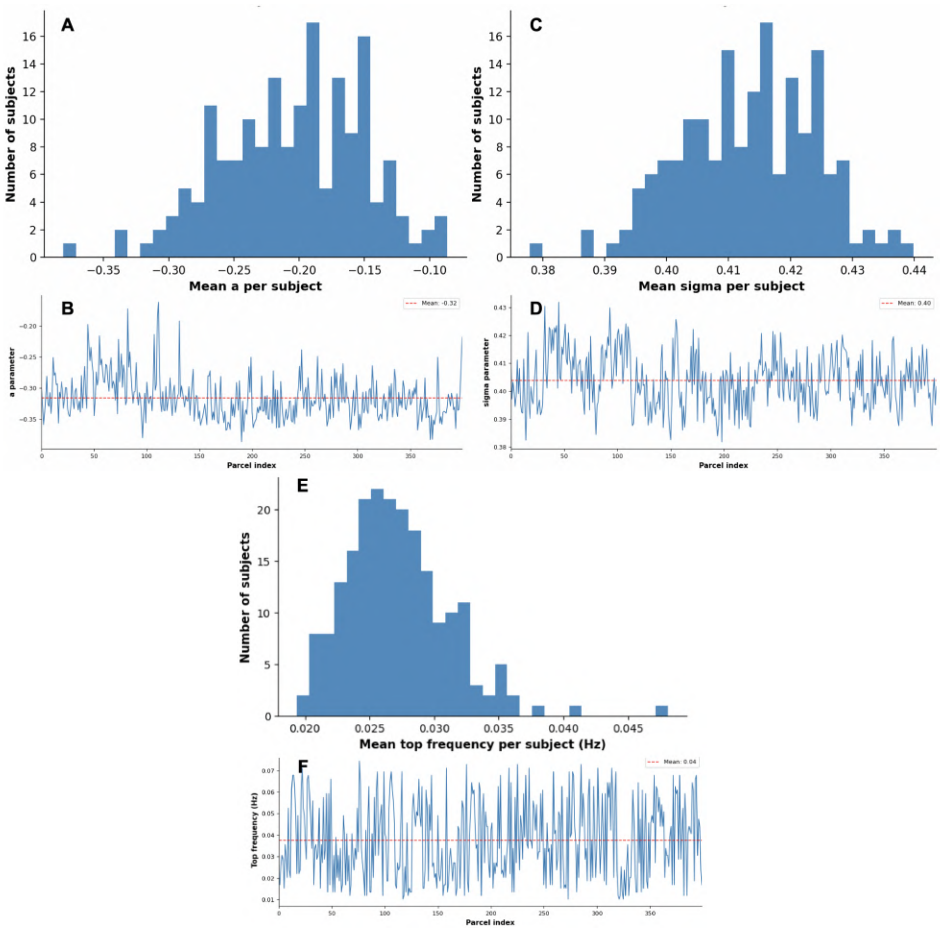

Figure A.3.3: Model parameters: bifurcation and noise parameters and top node frequency. In the histograms, we show the distributions of the subject mean (**A**) bifurcation parameter (a), (**C**) noise parameter (sigma), and (**E**) intrinsic frequency, respectively. These averages are computed over all nodes per subject. In the figures below the histograms, we show the respective distributions of one subject (**BDF**).

Code availability

The code to produce the results presented in this thesis is available at: `https://github.com/mellebolding/linhopf_fdt_adni`.

AI use statement

In the process of making this thesis, large language models were occasionally consulted to aid with coding and figure design. Additionally, occasional discussion with large language models contributed to locating literature and interpreting results. It must be noted that all such AI consultations were afterwards checked by me for truthfulness and consistency. All words in this thesis are my own unless otherwise specified.

A.4. Statistics

In this thesis, we use two different statistical tests: the Wilcoxon rank-sum test and the Monte-Carlo permutation test. In the case of independent quantities, we use the Wilcoxon rank-sum test. When this independency between measurements cannot be assumed, we make use of the Monte-Carlo permutation test. Here, we briefly describe the two tests.

Wilcoxon rank-sum test

The Wilcoxon rank-sum test is a non-parametric test used to assess whether two independent samples are drawn from the same distribution [76]. Given two independent samples $\{x_1, \dots, x_{n_1}\}$ and $\{y_1, \dots, y_{n_2}\}$, all observations are pooled and assigned ranks. The test statistic is then defined as the sum of ranks of one sample,

$$W = \sum_{i=1}^{n_1} R(x_i), \tag{A.4.0.1}$$

where $R(x_i)$ denotes the rank of observation x_i in the pooled sample. Under the null hypothesis that both samples are drawn from the same distribution, the distribution of W is known. For sufficiently large sample sizes, it is approximated by a normal distribution.

The corresponding z-score is computed and the p-value is obtained from the standard normal distribution. In this thesis, the Wilcoxon rank-sum test is applied to subject-averaged quantities, for which independence between observations can be reasonably assumed.

Monte-Carlo permutation test

The Monte-Carlo permutation test is a non-parametric resampling-based test that assesses statistical significance without assuming independence or a specific underlying distribution [77]. Let T_{obs} denote the observed value of a test statistic (e.g. a difference in means) computed from the original data. To construct the null distribution, labels are randomly permuted N_{perm} times while preserving the structure of the data. Subsequently, for each permutation k, the test statistic T_k is computed.

The p-value is estimated as,

$$p = \frac{1 + \sum_{k=1}^{N_{\text{perm}}} \mathbb{I}(|T_k| \geq |T_{\text{obs}}|)}{1 + N_{\text{perm}}}, \tag{A.4.0.2}$$

where $\mathbb{I}(\cdot)$ is the indicator function. This formulation ensures an unbiased p-value estimate for even a finite numbers of permutations. In this thesis, the Monte-Carlo permutation test is applied to parcel-averaged quantities, where spatial or network-level dependencies violate independence assumptions.

A.5. Duhamel derivation

We solve the N-dimensional Langevin equation, Eq. 3.3.0.1,

$$\frac{d\mathbf{u}(t)}{dt} = -\mathbf{\Gamma}\,\mathbf{u}(t) + \boldsymbol{\eta}(t), \qquad \mathbf{u}(0) = \mathbf{u}_0, \tag{A.5.0.1}$$

where $\mathbf{\Gamma} \in \mathbb{R}^{N\times N}$ is constant and $\boldsymbol{\eta}(t)$ is Gaussian white noise.

Introduce the state-transition matrix,

$$\Phi(t) \equiv e^{-\mathbf{\Gamma}t},$$

which satisfies $\dot{\Phi}(t) = -\mathbf{\Gamma}\,\Phi(t)$ and $\Phi(0) = \mathbf{I}$. Multiplying (A.5.0.1) on the left by $\Phi(t)$ and using the product rule gives,

$$\frac{d}{dt}\big[\Phi(t)\mathbf{u}(t)\big] = \dot{\Phi}(t)\mathbf{u}(t) + \Phi(t)\dot{\mathbf{u}}(t) = -\mathbf{\Gamma}\Phi(t)\mathbf{u}(t) + \Phi(t)\big(-\mathbf{\Gamma}\mathbf{u}(t) + \boldsymbol{\eta}(t)\big) = \Phi(t)\boldsymbol{\eta}(t).$$

Integrating from 0 to t and making use of $\Phi(0) = \mathbf{I}$ yields the variation-of-constants (Duhamel) formula,

$$\Phi(t)\mathbf{u}(t) - \mathbf{u}_0 = \int_0^t \Phi(\tau)\,\boldsymbol{\eta}(\tau)\,d\tau.$$

Multiplying on the left by $\Phi(t)^{-1} = e^{\mathbf{\Gamma}t}$ gives the solution,

$$\mathbf{u}(t) = e^{-\mathbf{\Gamma}t}\,\mathbf{u}_0 \;+\; \int_0^t e^{-\mathbf{\Gamma}(t-\tau)}\,\boldsymbol{\eta}(\tau)\,d\tau \tag{A.5.0.2}$$

which gives Eq. 3.3.0.3 in Sec. 3.3.

A.6. Additional figures

In this section of the appendix we show some figures that are not as relevant for the main narrative of the thesis. In Fig. A.6.1, we show the normalized changes of average parcel values of the HC to AD groups for the tau, amyloid-beta, and both IV metrics. Meanwhile, in Fig. A.6.2, the IV subject averages of Fig. 4.2.4**A** and 4.2.4**B** are shown in a radar plot format. Here, the MCI values are added as well. We can see that for both the model-free and analytical IV approach, the distributions are similar. Therefore, we do not expect and do not find different results between MCI and HC/AD.

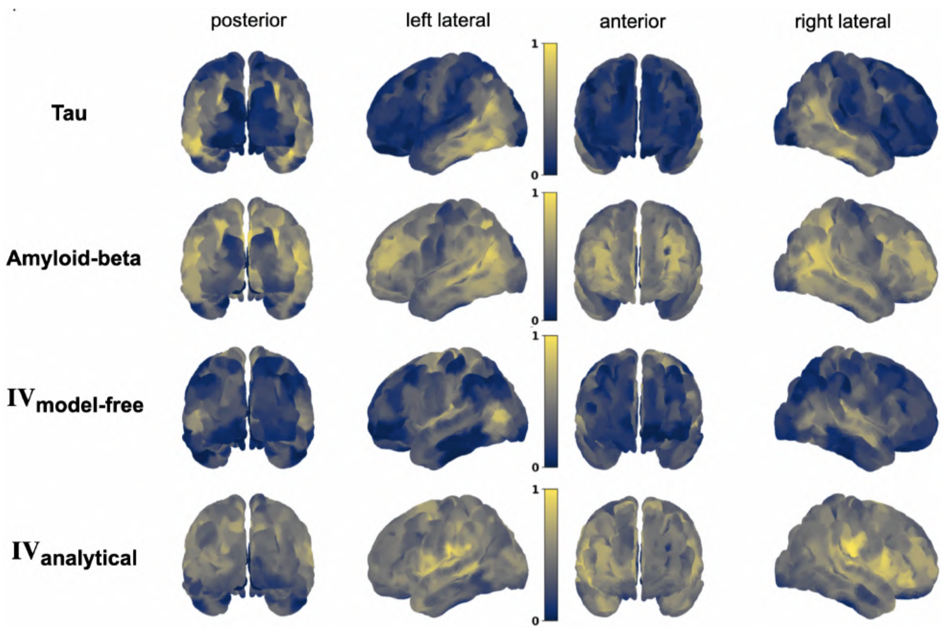

Figure A.6.1: Normalized changes of average parcel values of the HC to AD groups for tau, amyloid-beta, and both IV metrics. The regions in yellow indicate those regions with the largest change in value when going from HC to AD. The plots are normalized per metric type.

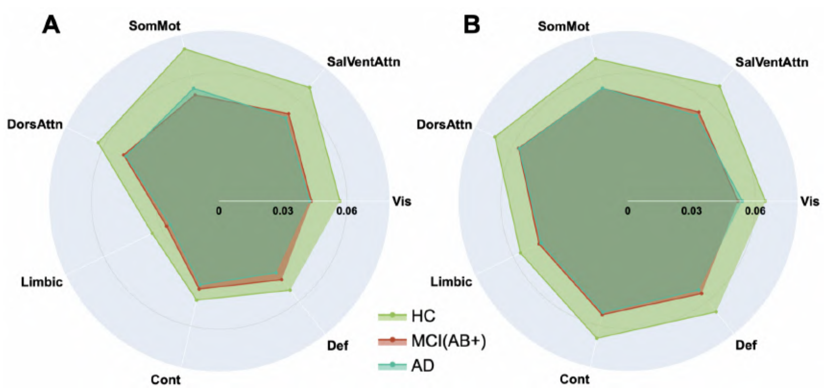

Figure A.6.2: Integral violation and resting-state networks. Radar plots of the subject averages of the integral violation of each RSN for each group (HC, MCI, AD) for both the model-free (**A**) and analytical (**B**) approach.

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